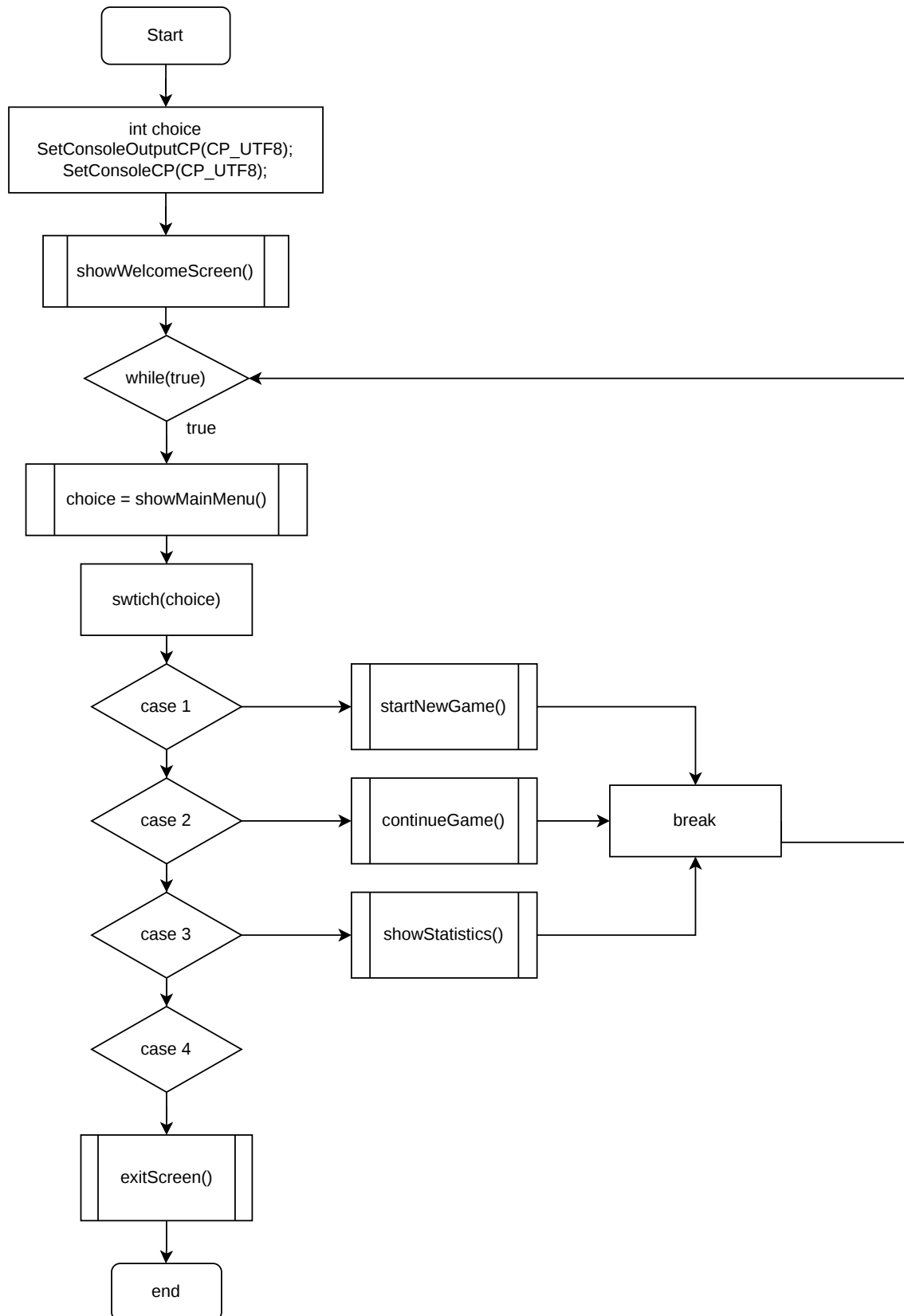


5.0 Flowchart

5.1 Chess

5.1.1 int main

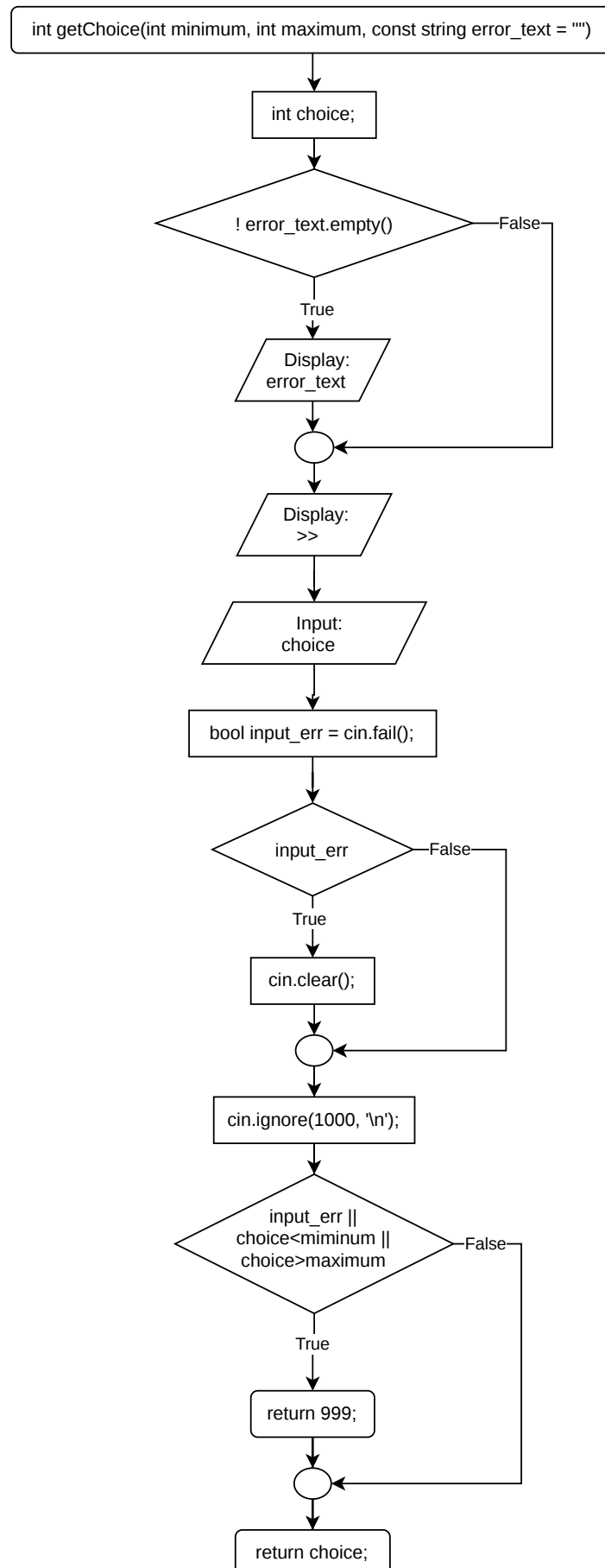
The main() function controls the overall execution of the Console Chess Game.



5.2 Game Flow

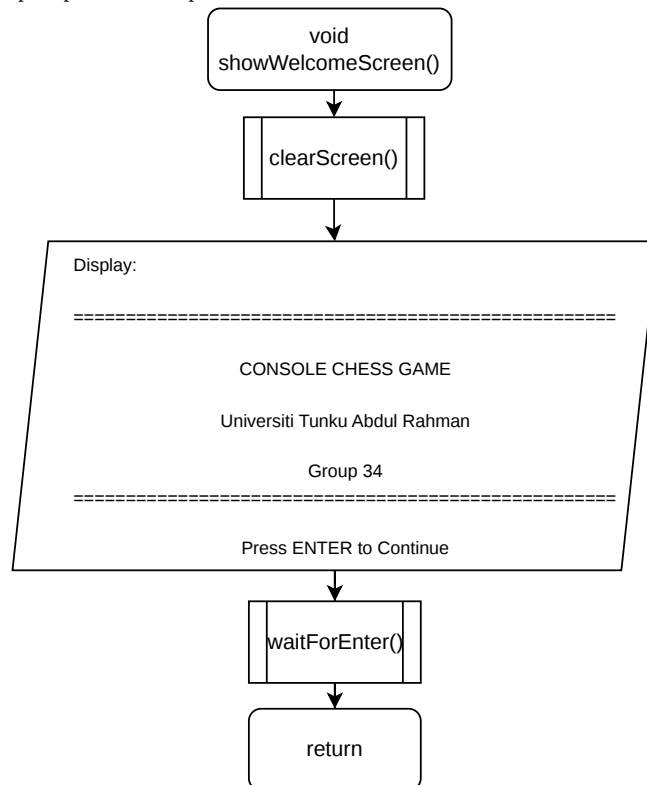
5.2.1 getChoice

This function reads and validates a numeric menu choice within a specified range, and returns if the input is invalid.



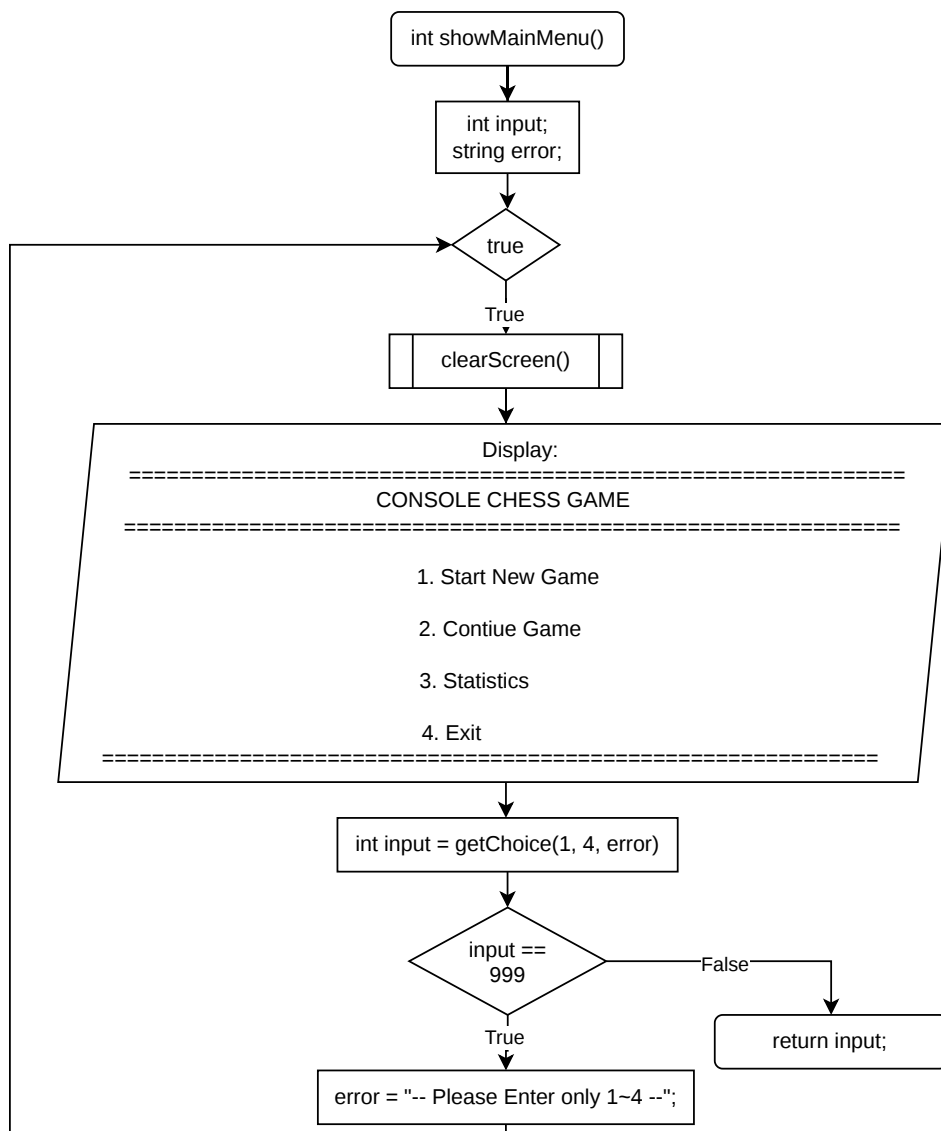
5.2.2 showWelcomeScreen

This function displays the welcome screen and prompts the user to press enter to continue.



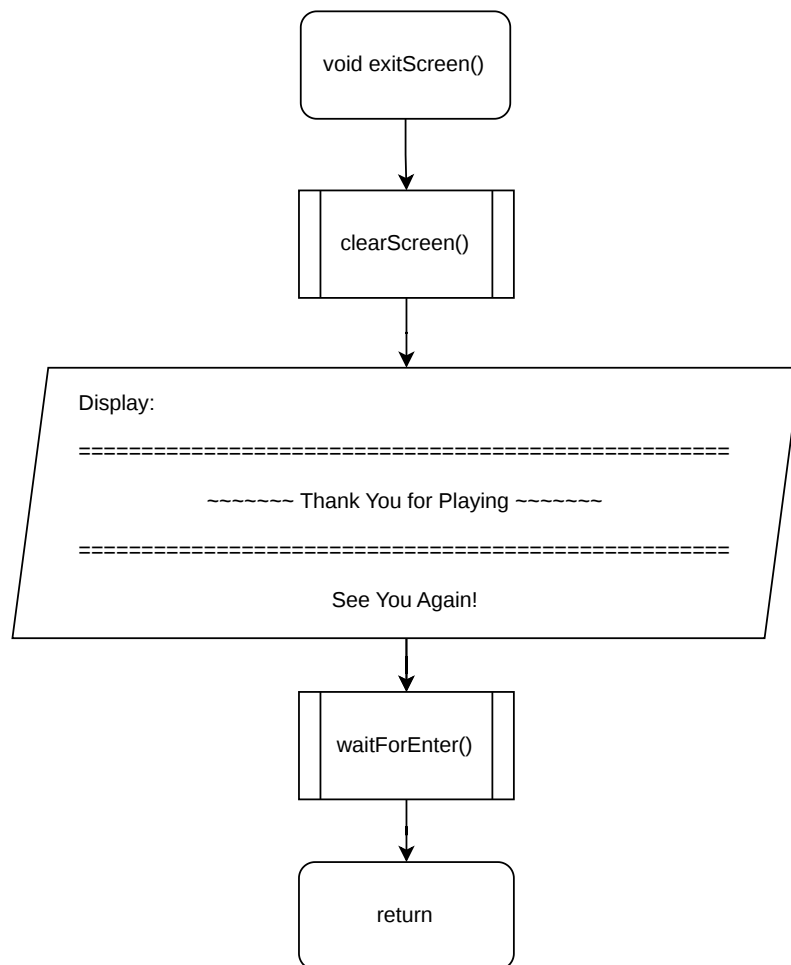
5.2.3 showMainMenu

This function displays the main menu options and returns the user's valid input of choice.



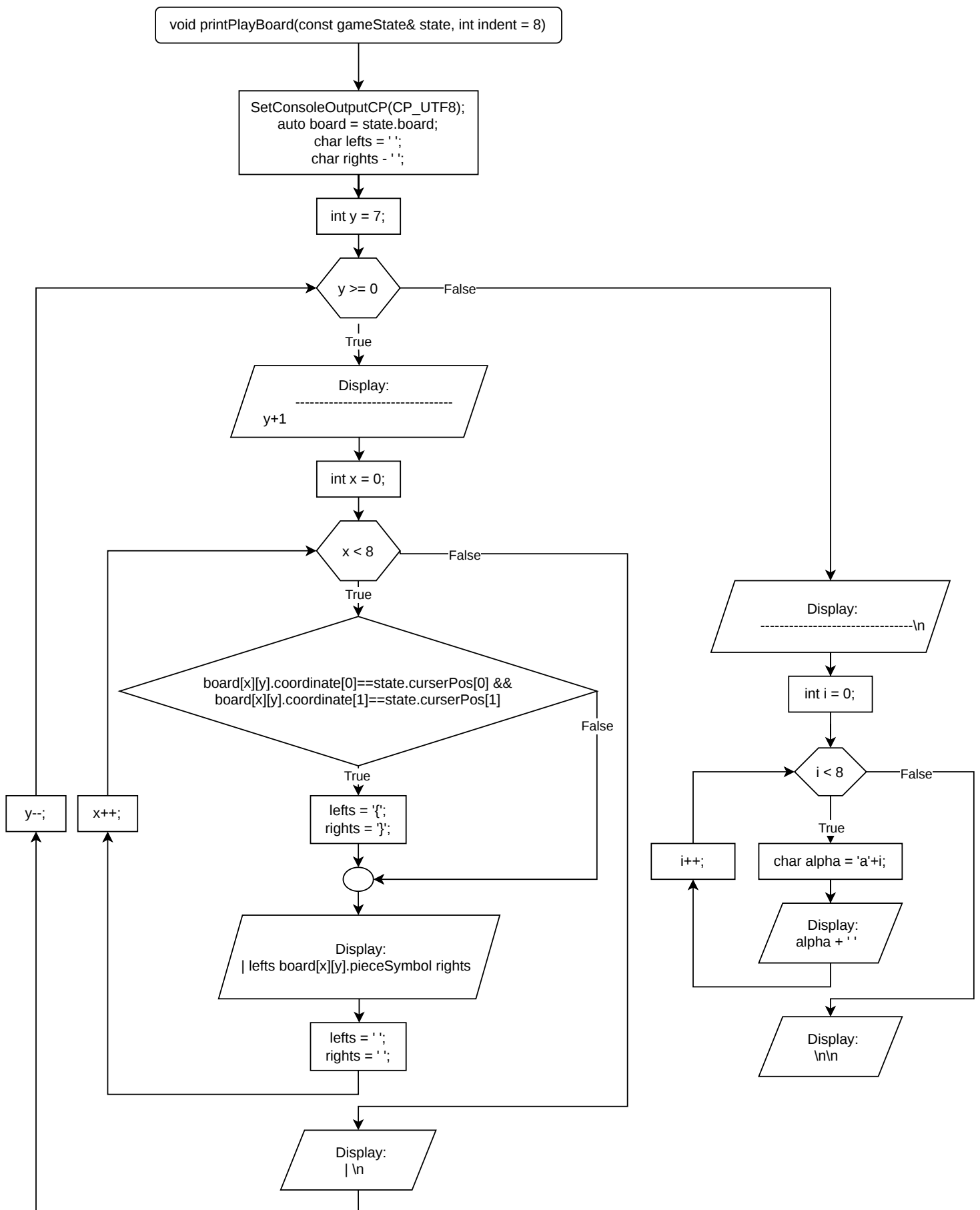
5.2.4 exitScreen

This function displays a thank you message before user exits the application.



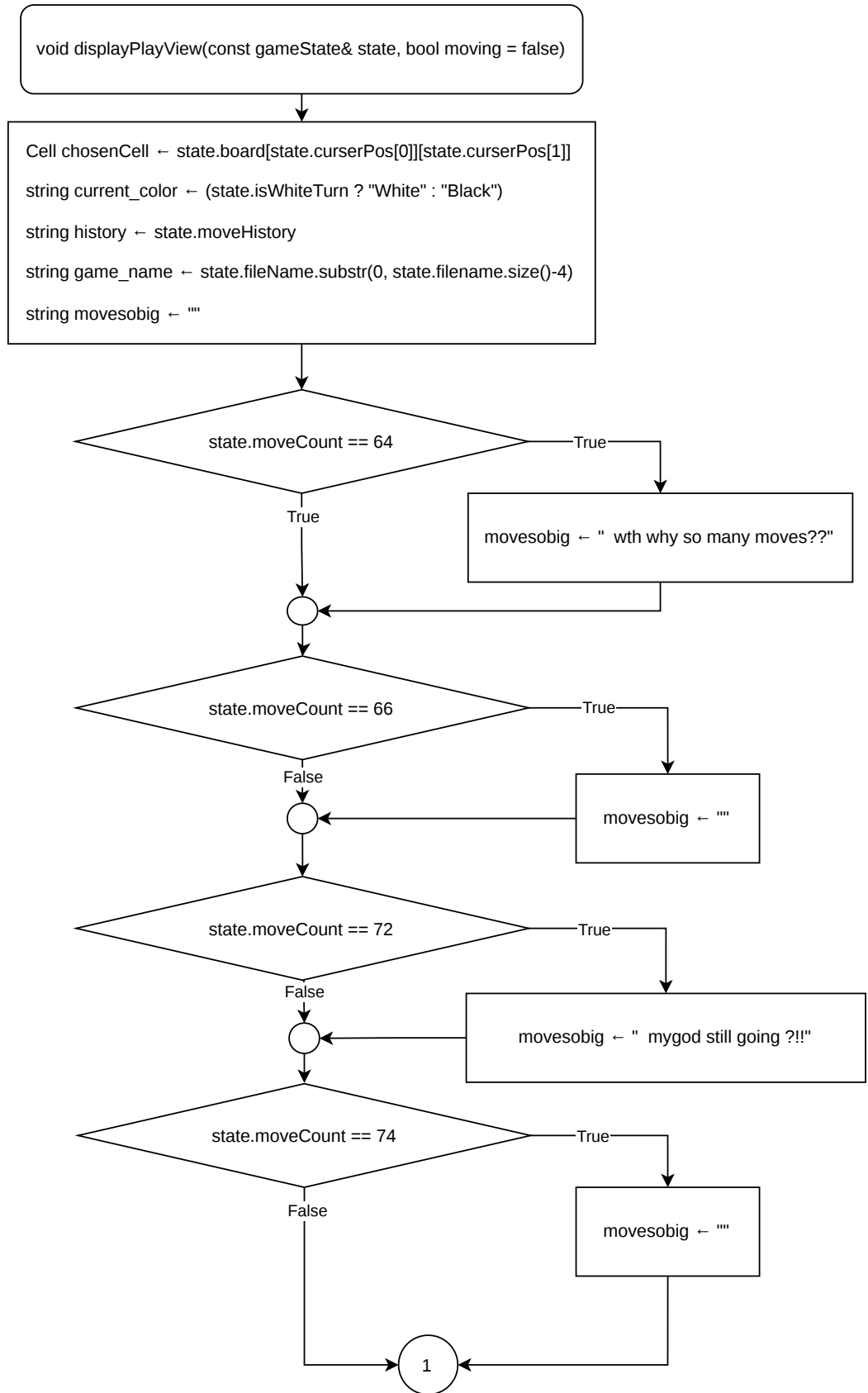
5.2.5 printPlayBoard

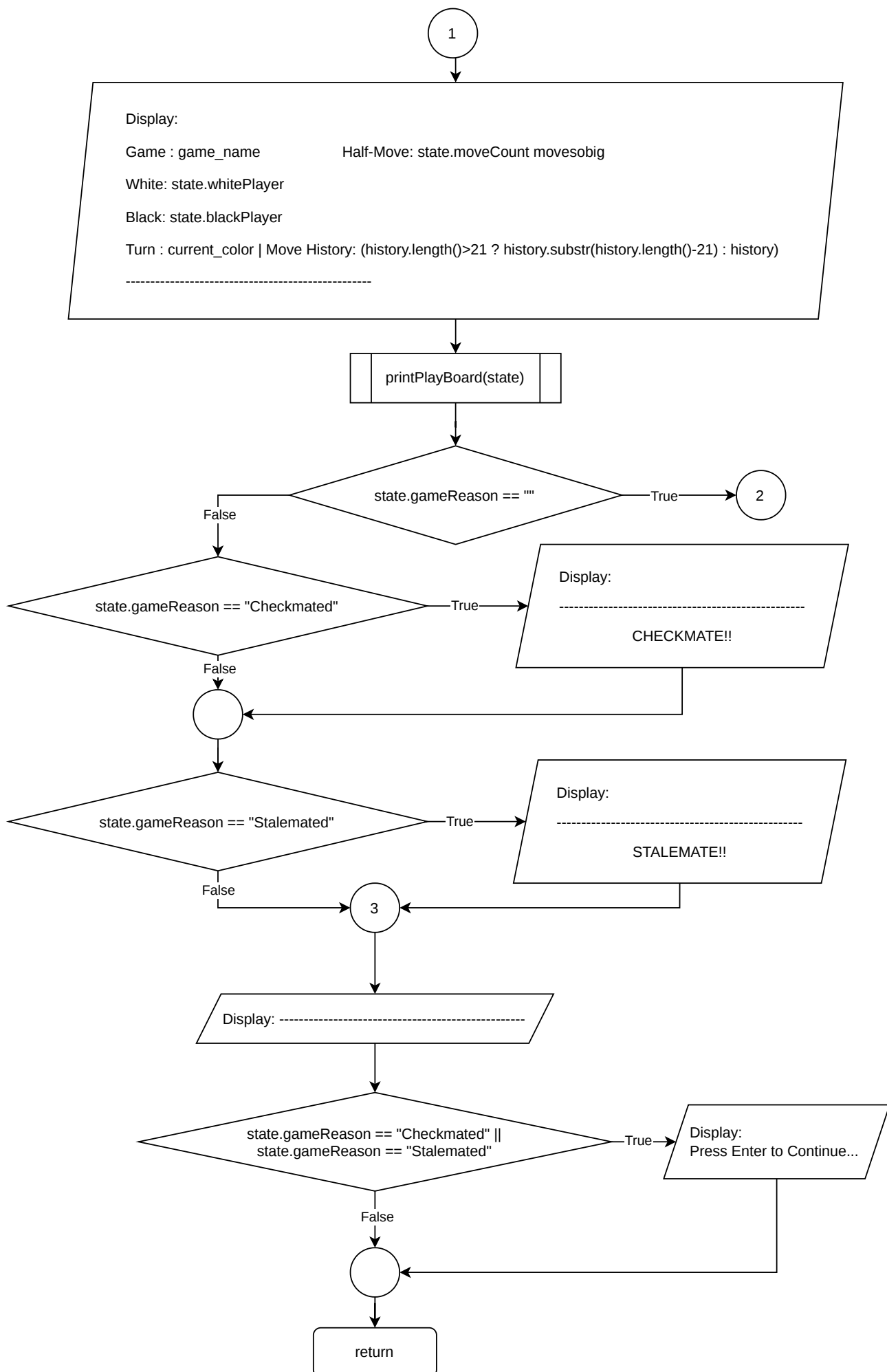
This function displays the 8 x 8 chess board matrix with coordinate labels and selection cursor.

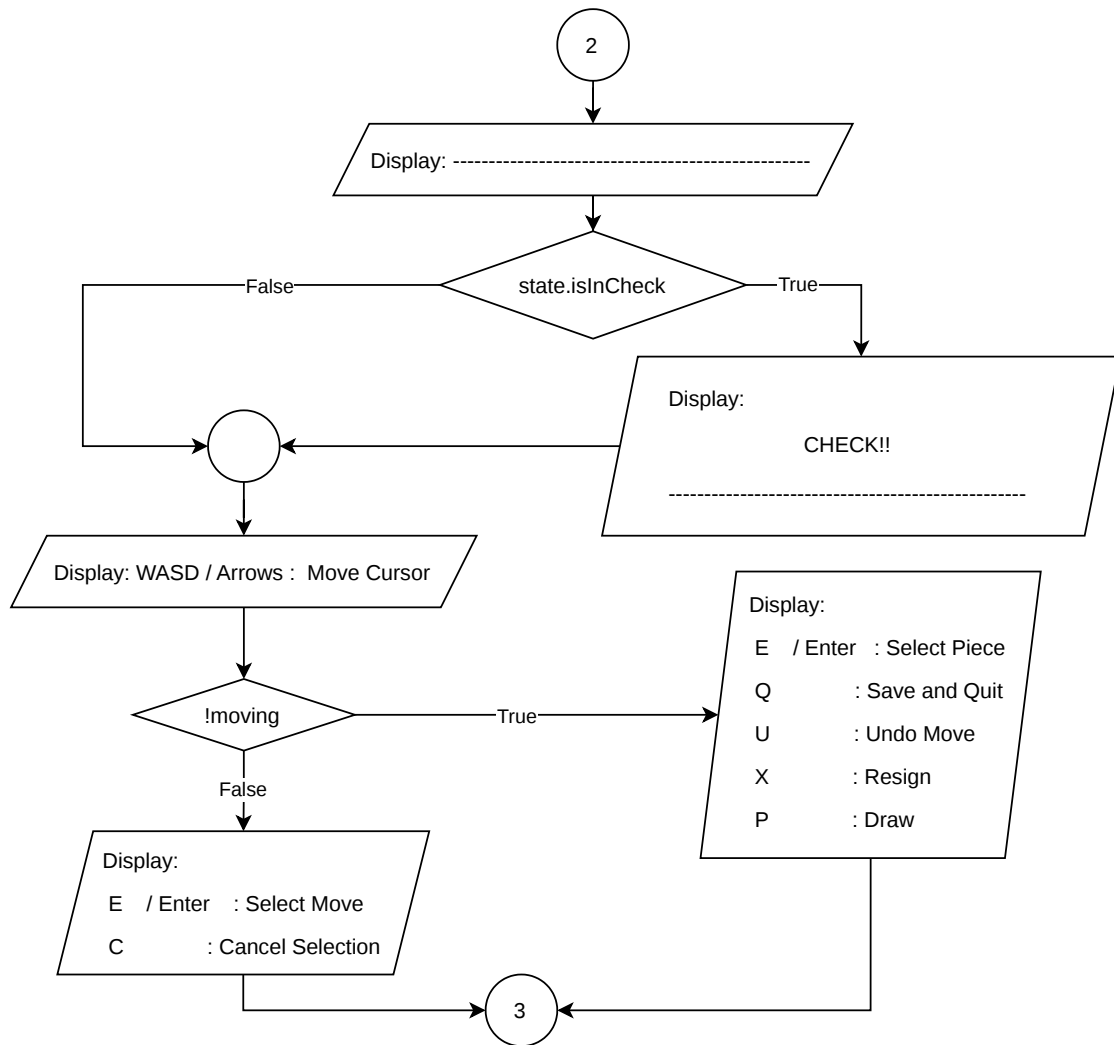


5.2.6 displayPlayView

This function displays the 8 x 8 chess board matrix with coordinate labels and selection cursor.

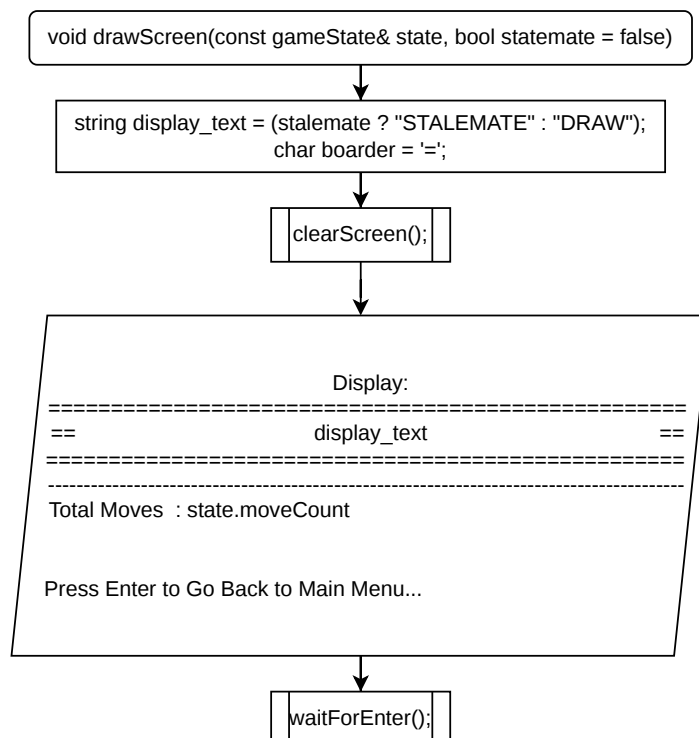






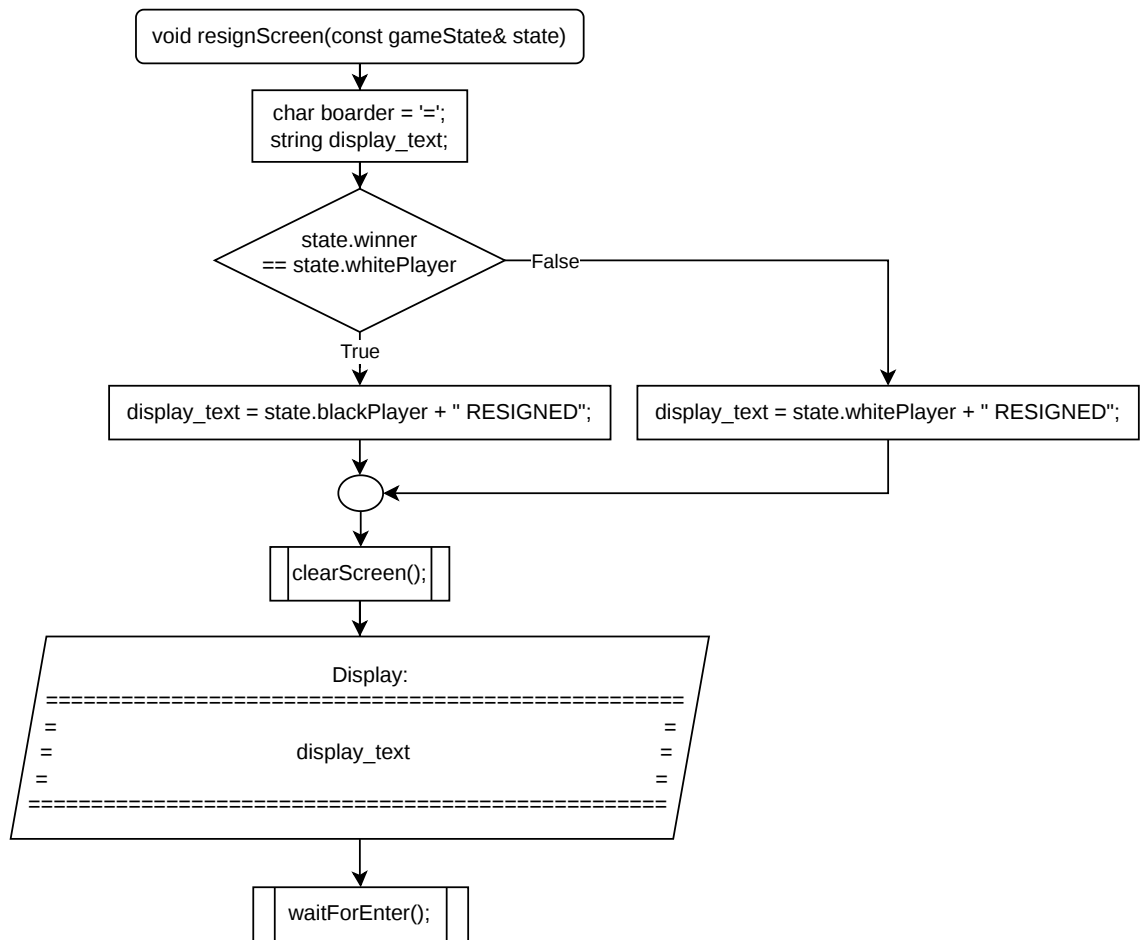
5.2.7 drawScreen

This function displays the match summary screen when a game ends in mutual draw or stalemate.



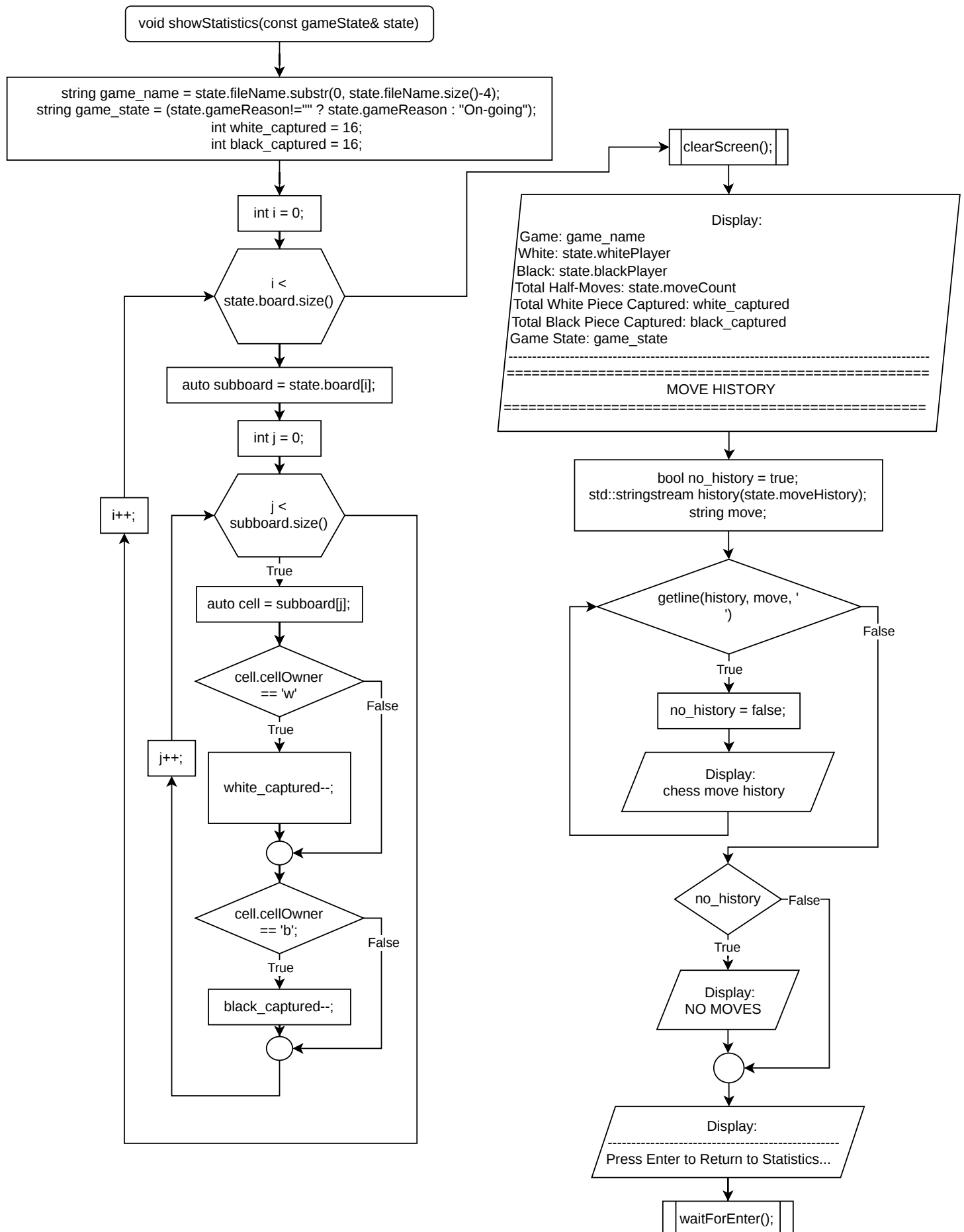
5.2.8 resignScreen

This function displays a resignation notice when a party surrenders.



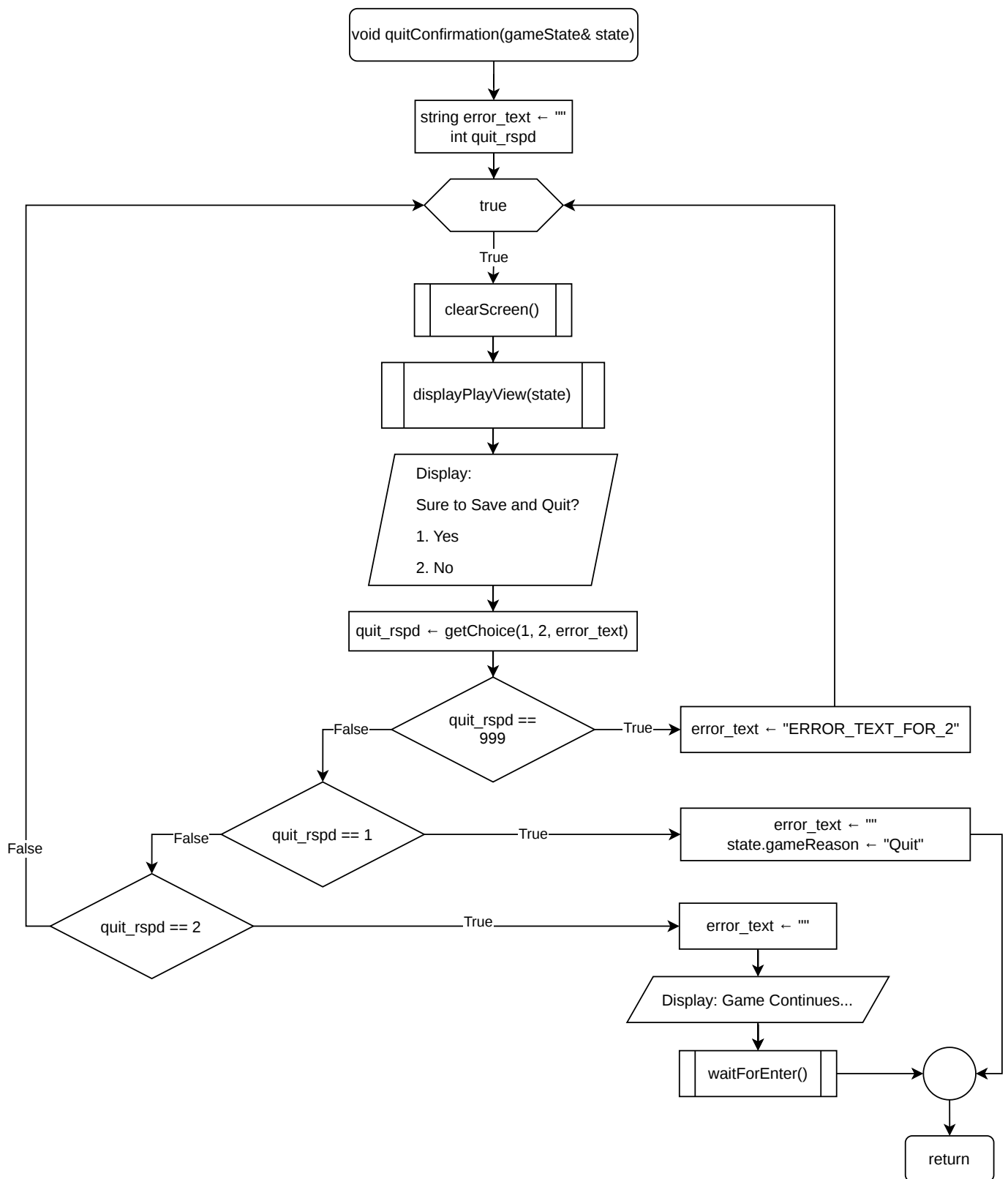
5.2.9 showStatistics

This function displays detailed match statistics for a match played including move history and capture tallies.



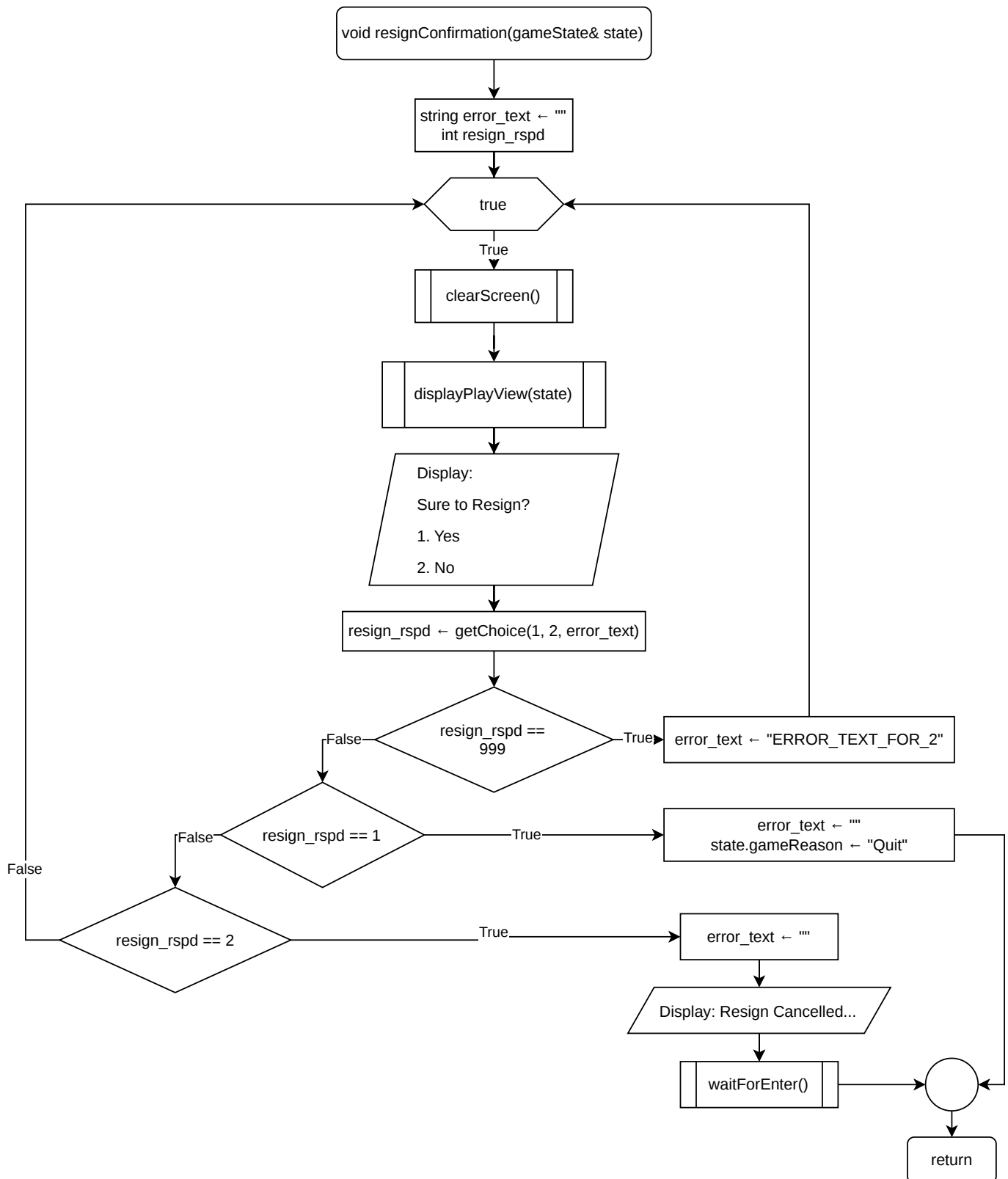
5.2.10 quitConfirmation

This function prompts the active player to confirm saving and quitting the game, and return to the main menu if confirmed.



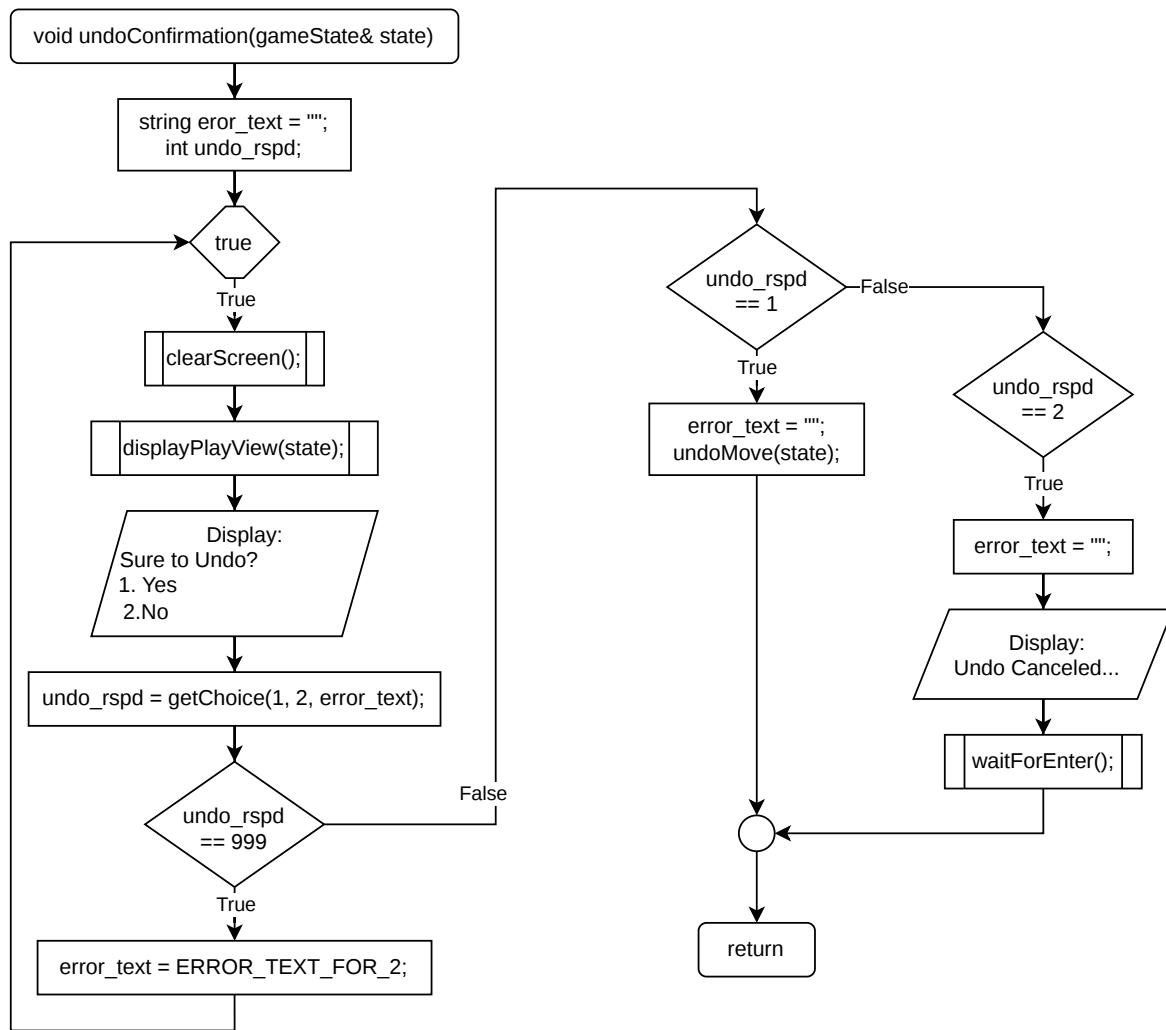
5.2.11 resignConfirmation

This function asks the player's confirmation to resign the match and awards the win to the component if confirmed.



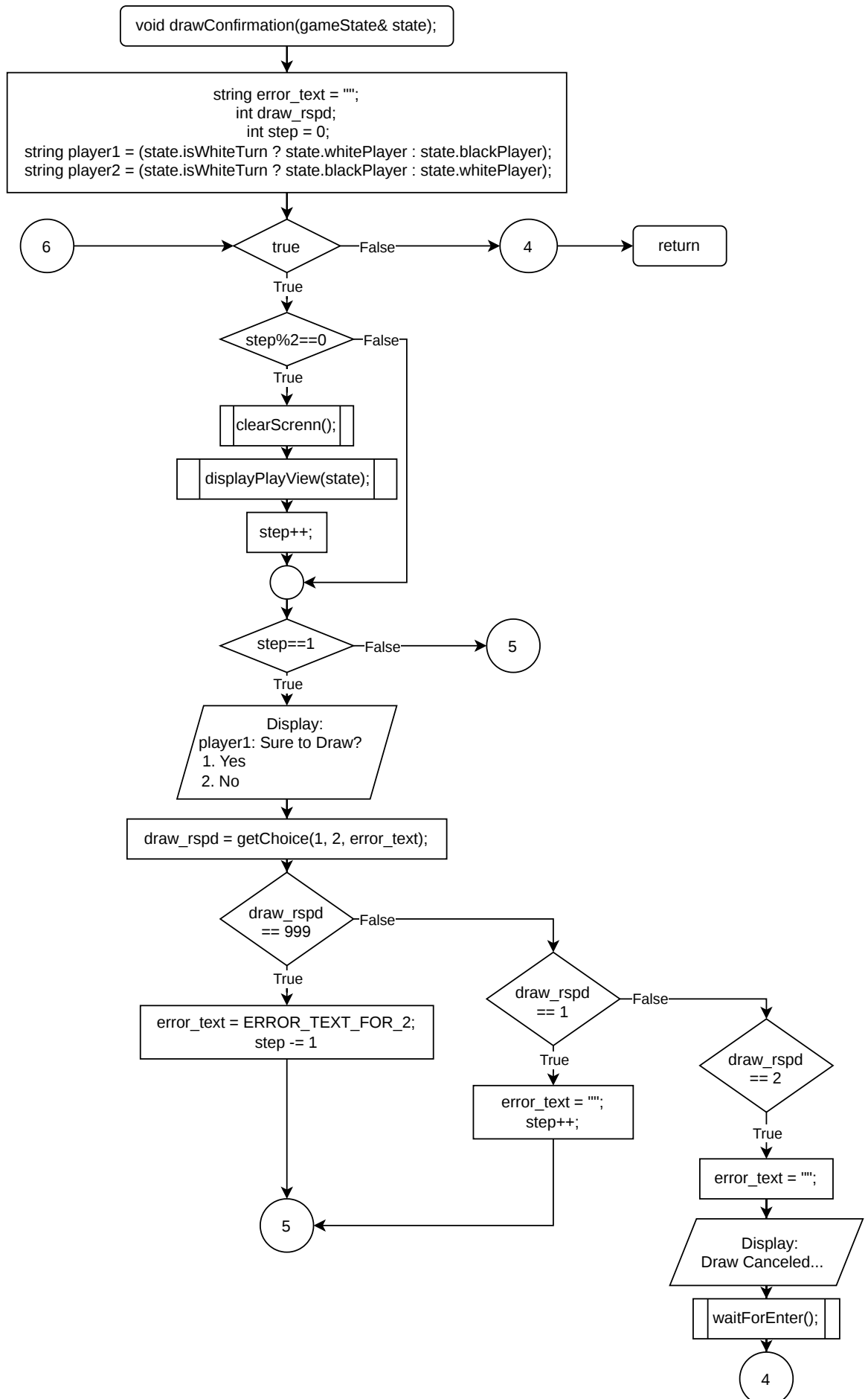
5.2.12 undoConfirmation

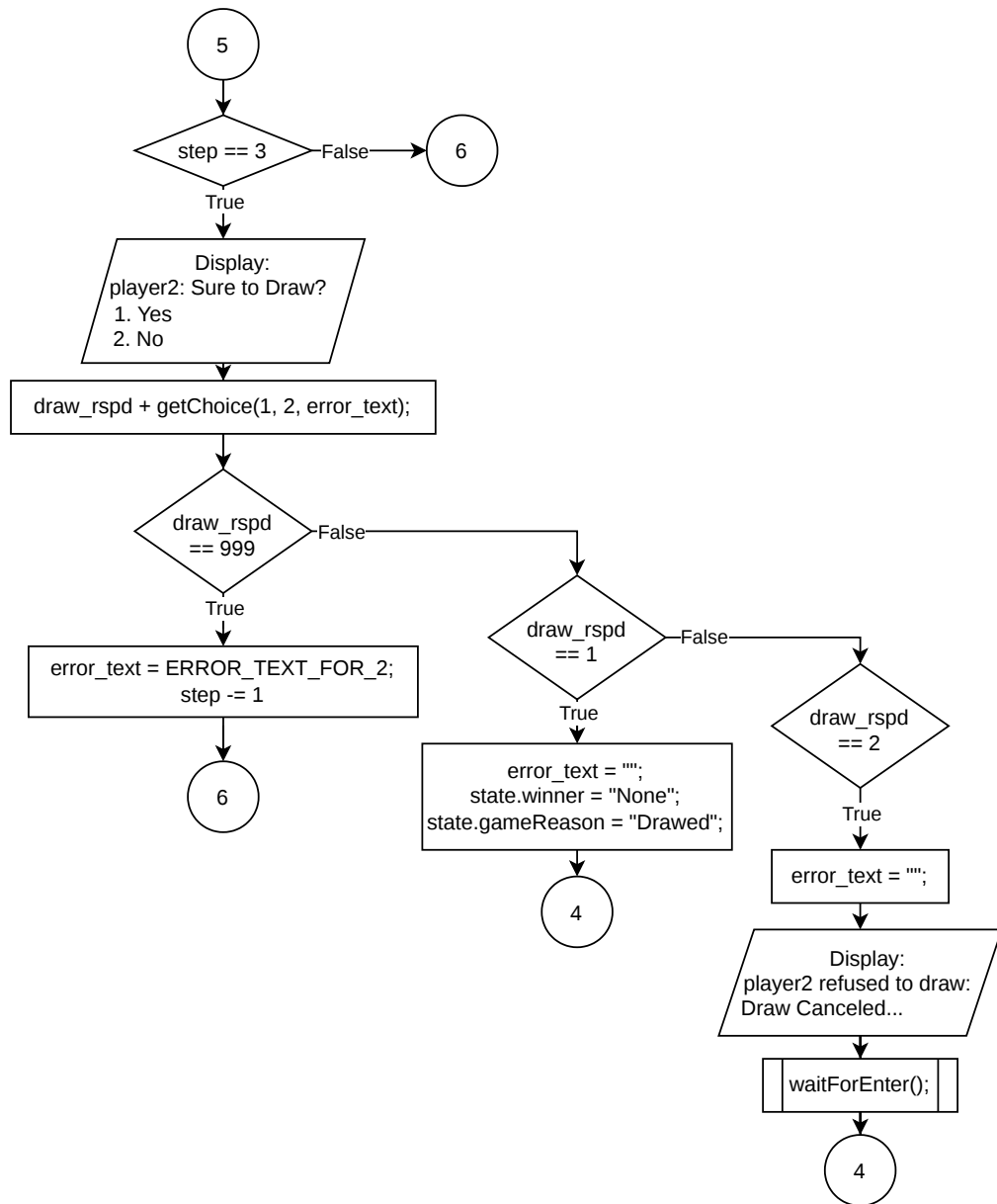
This function prompts the active player to confirm saving and quitting the game, and return to the main menu if confirmed.



5.2.13 drawConfirmation

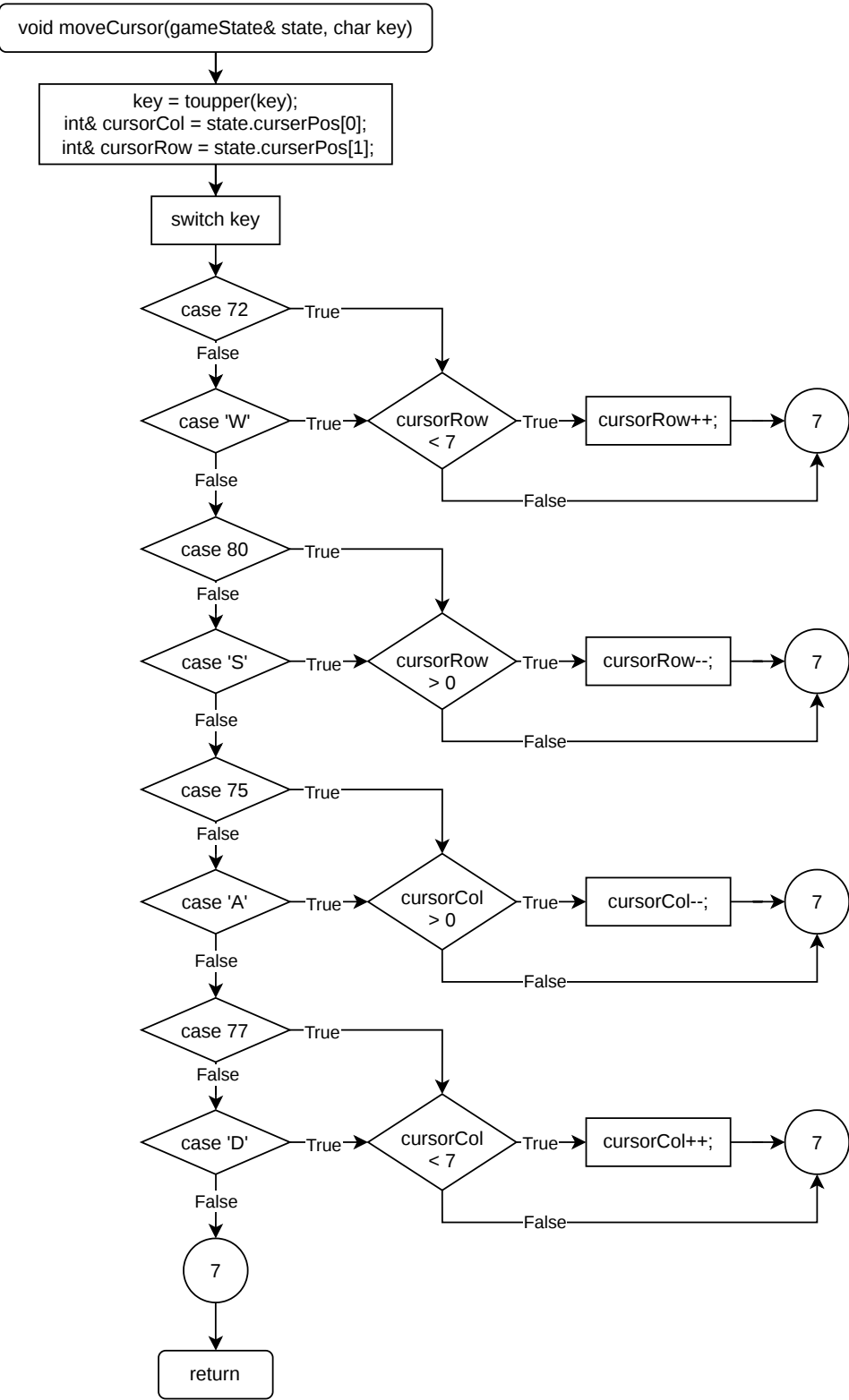
This function prompts both players to agree on resignation upon called.





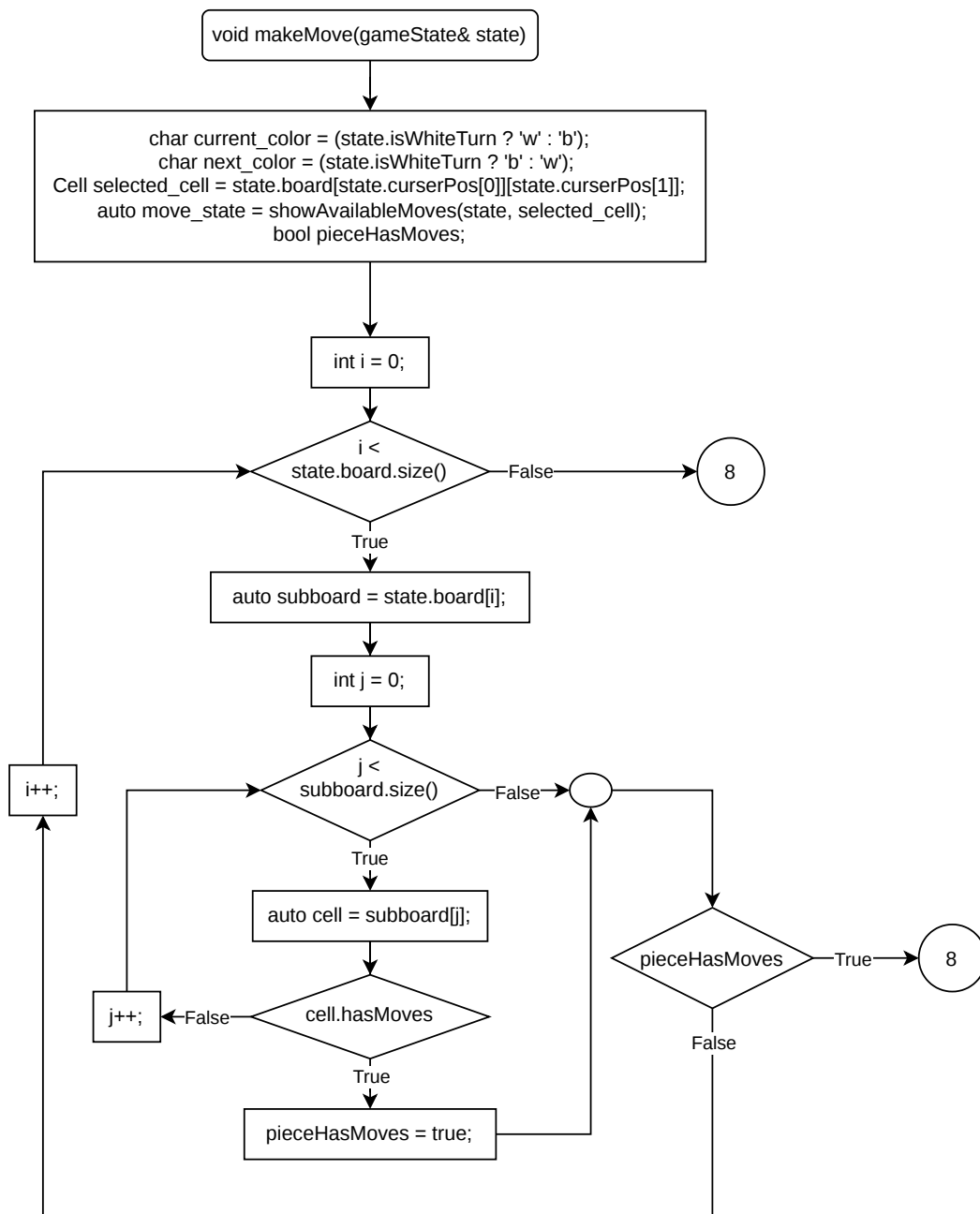
5.2.14 moveCursor

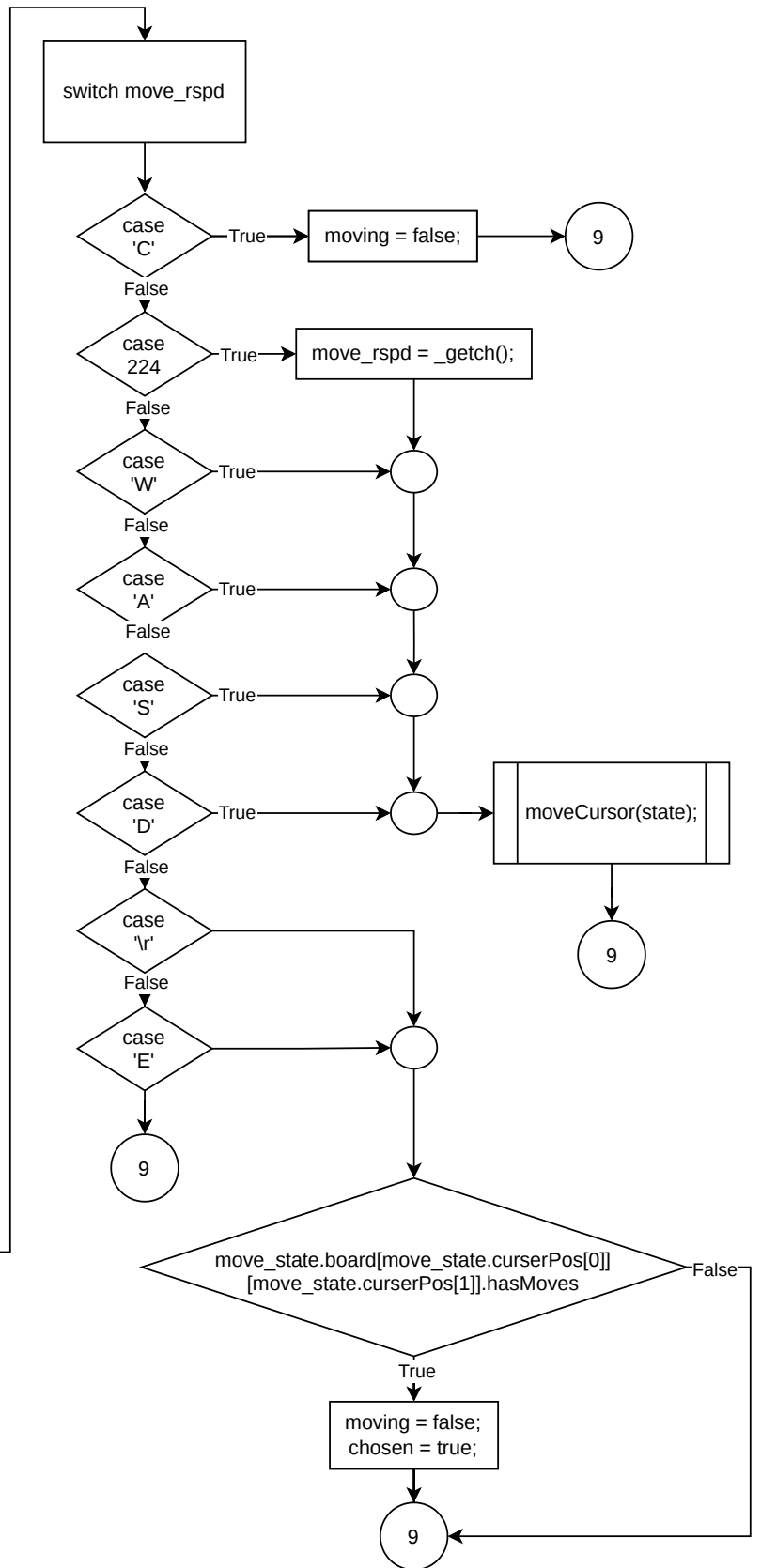
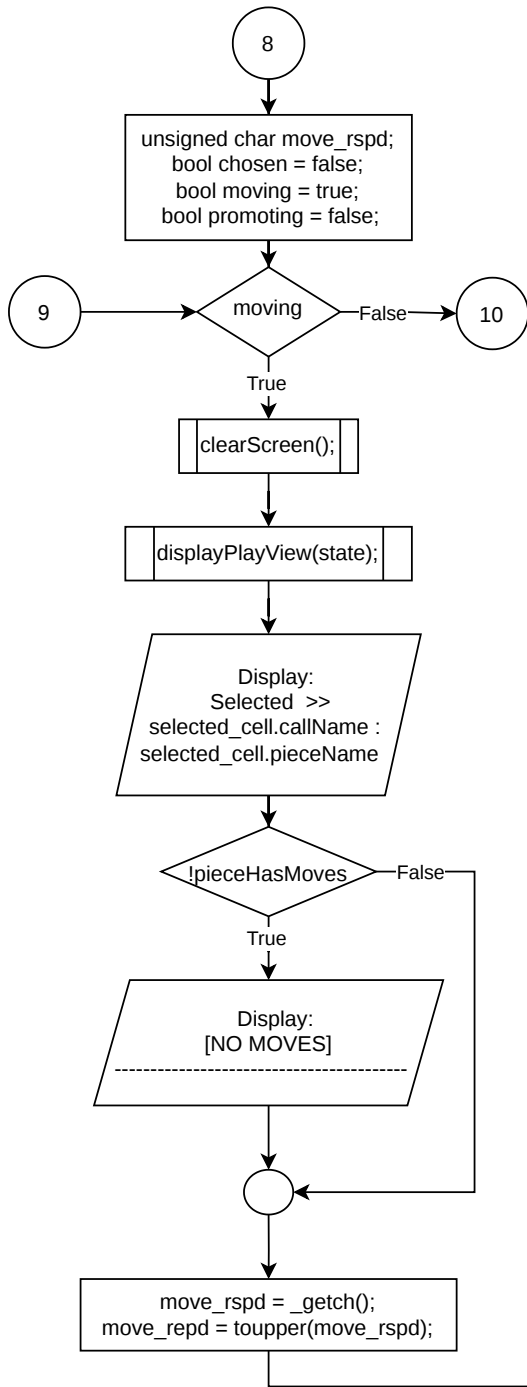
This function moves the cursor on the board when WASD or directional keys are pressed.

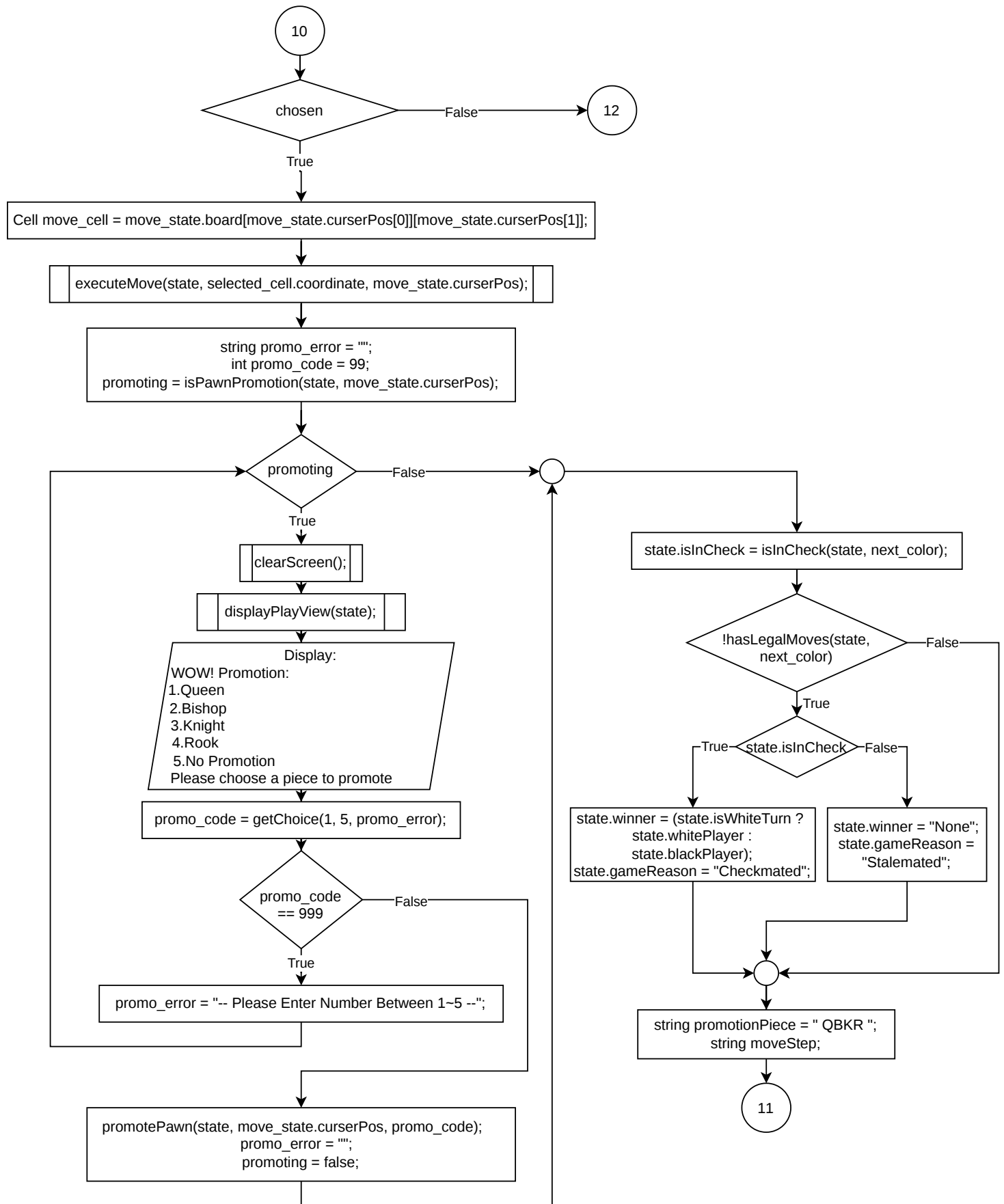


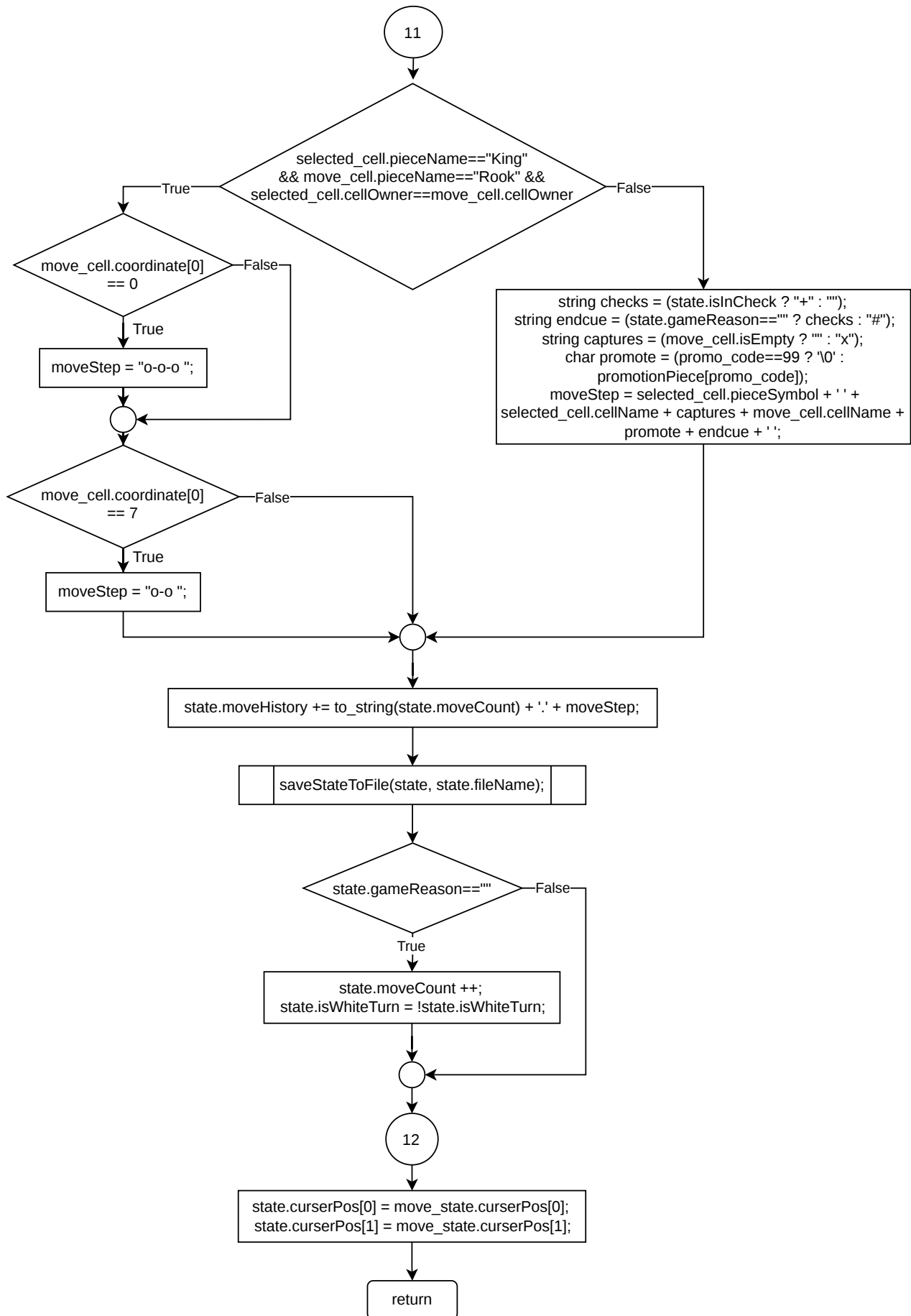
5.2.15 makeMove

This function moves the cursor on the board when WASD or directional keys are pressed.



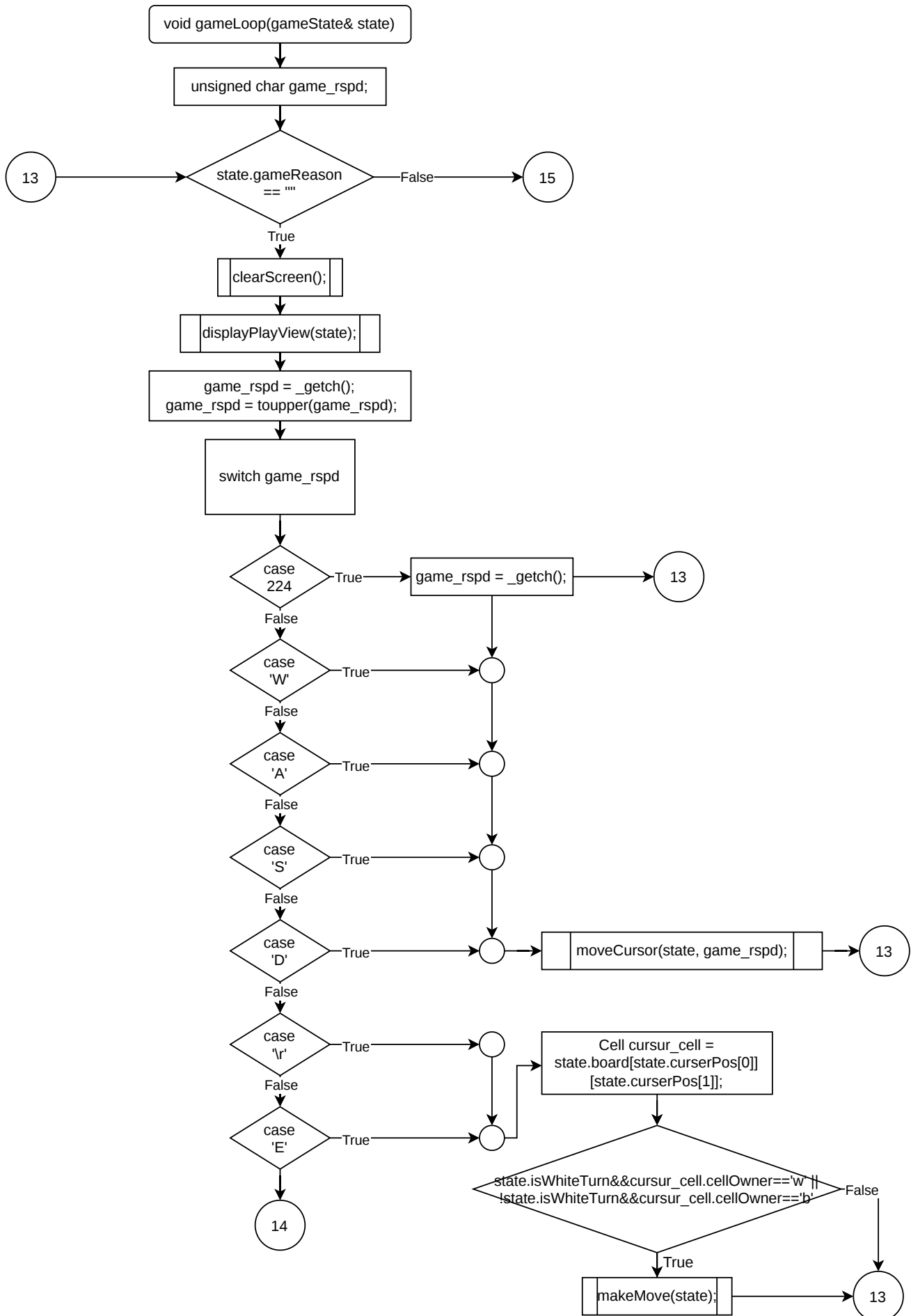


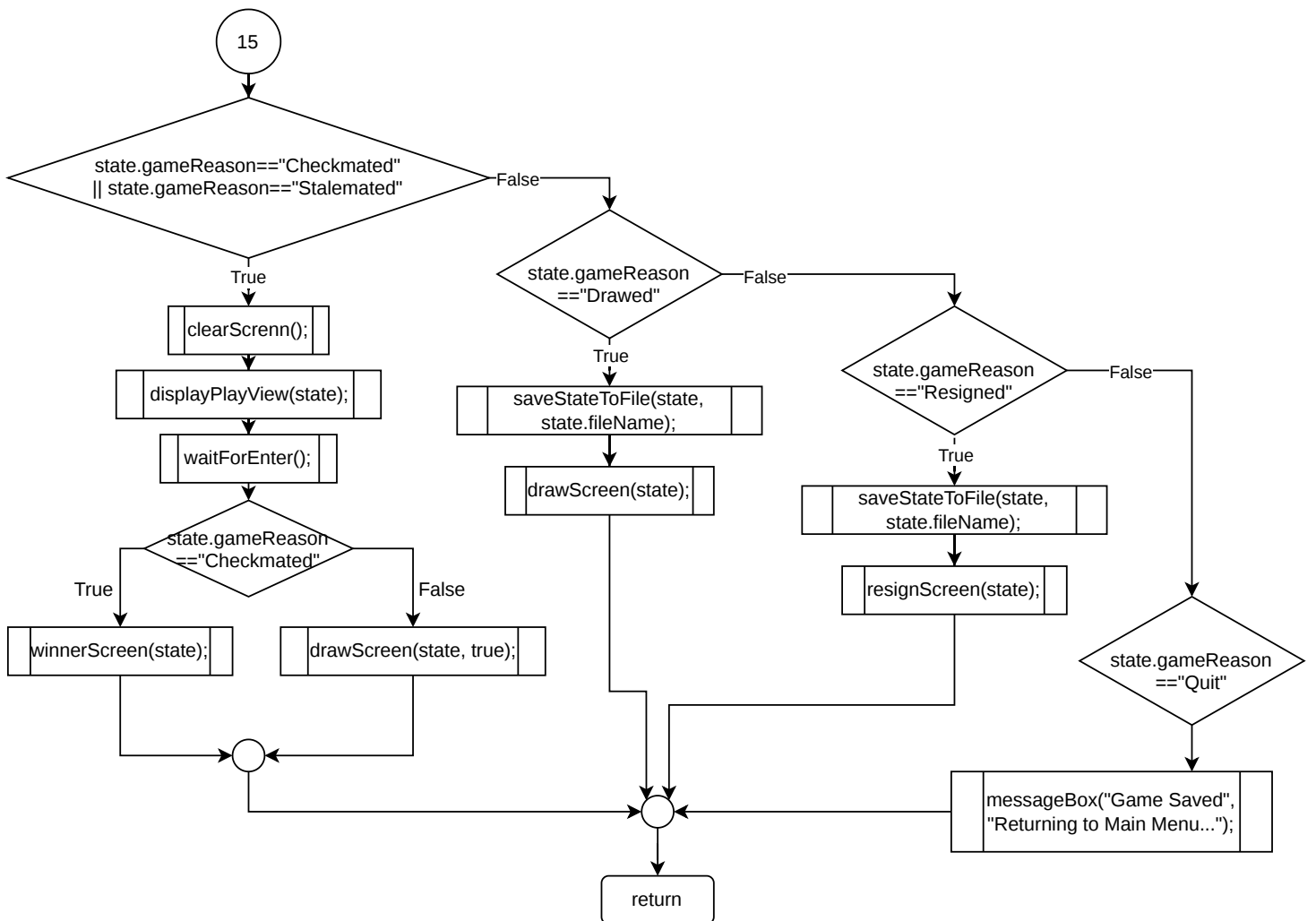
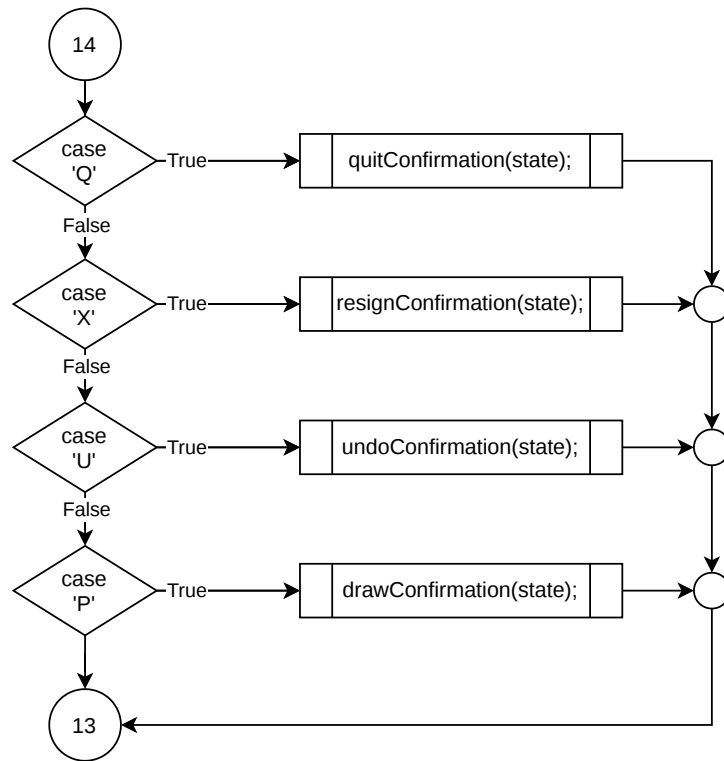




5.2.16 gameLoop

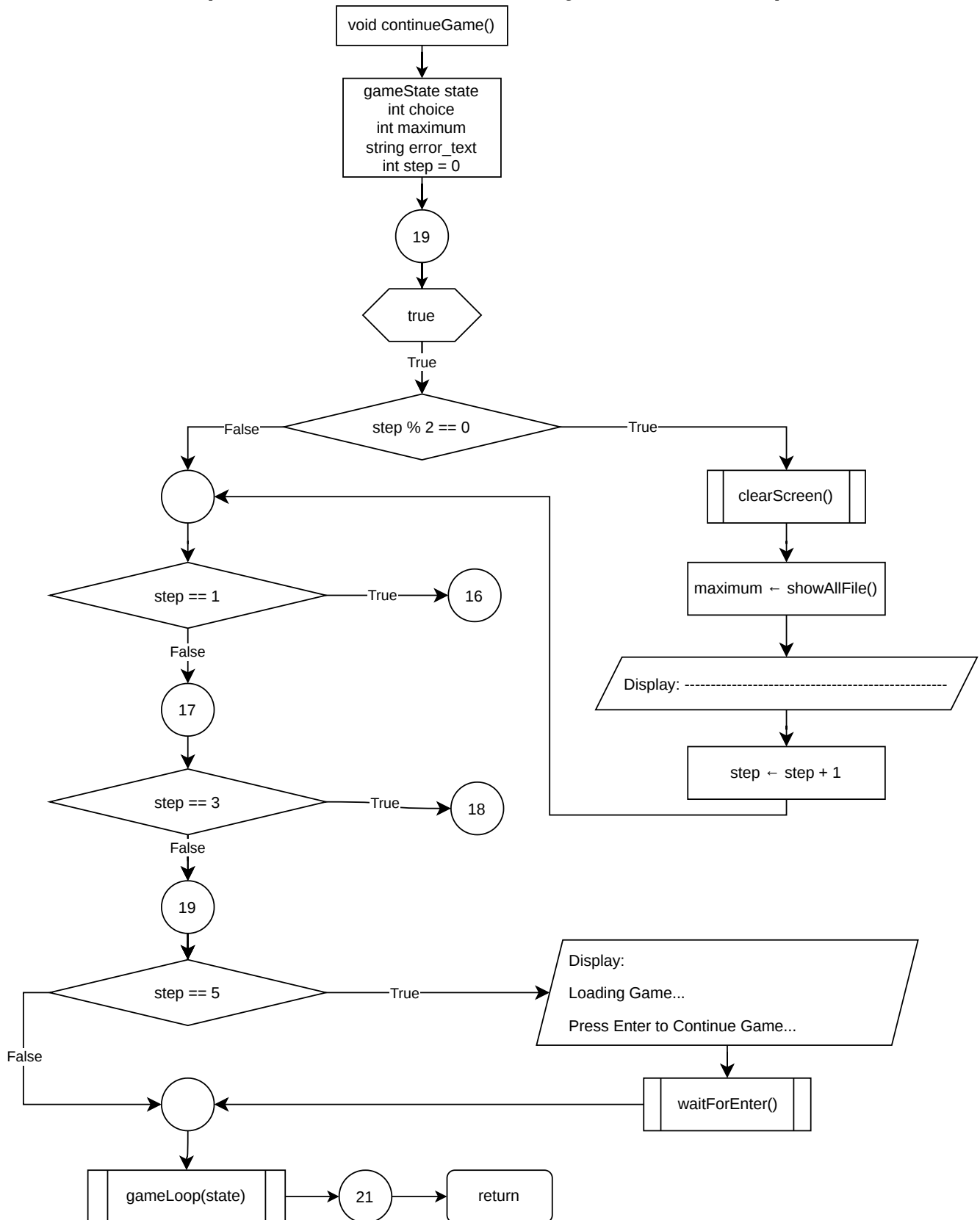
This function is the main game loop that constantly and actively receives player input during the game.

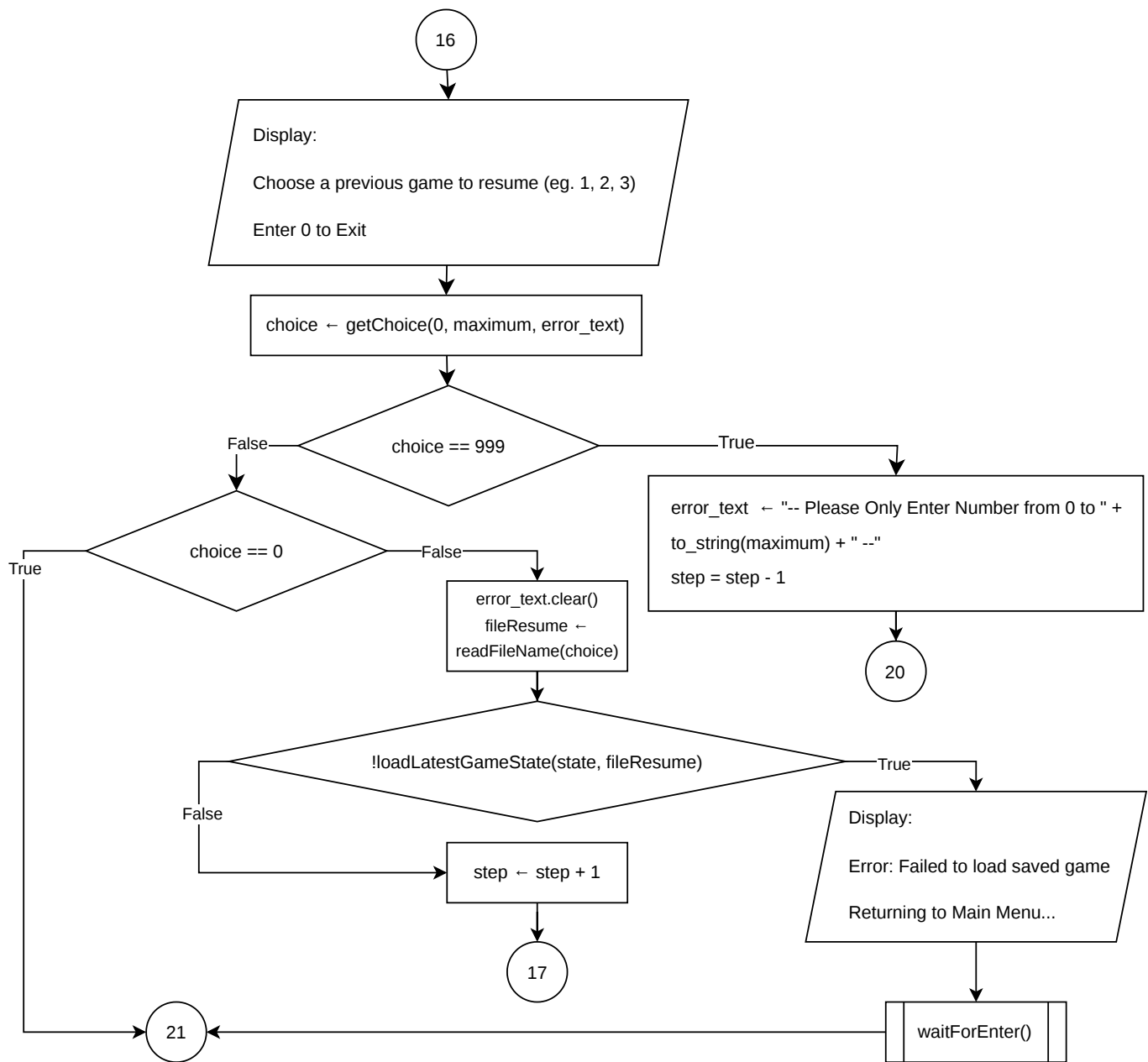


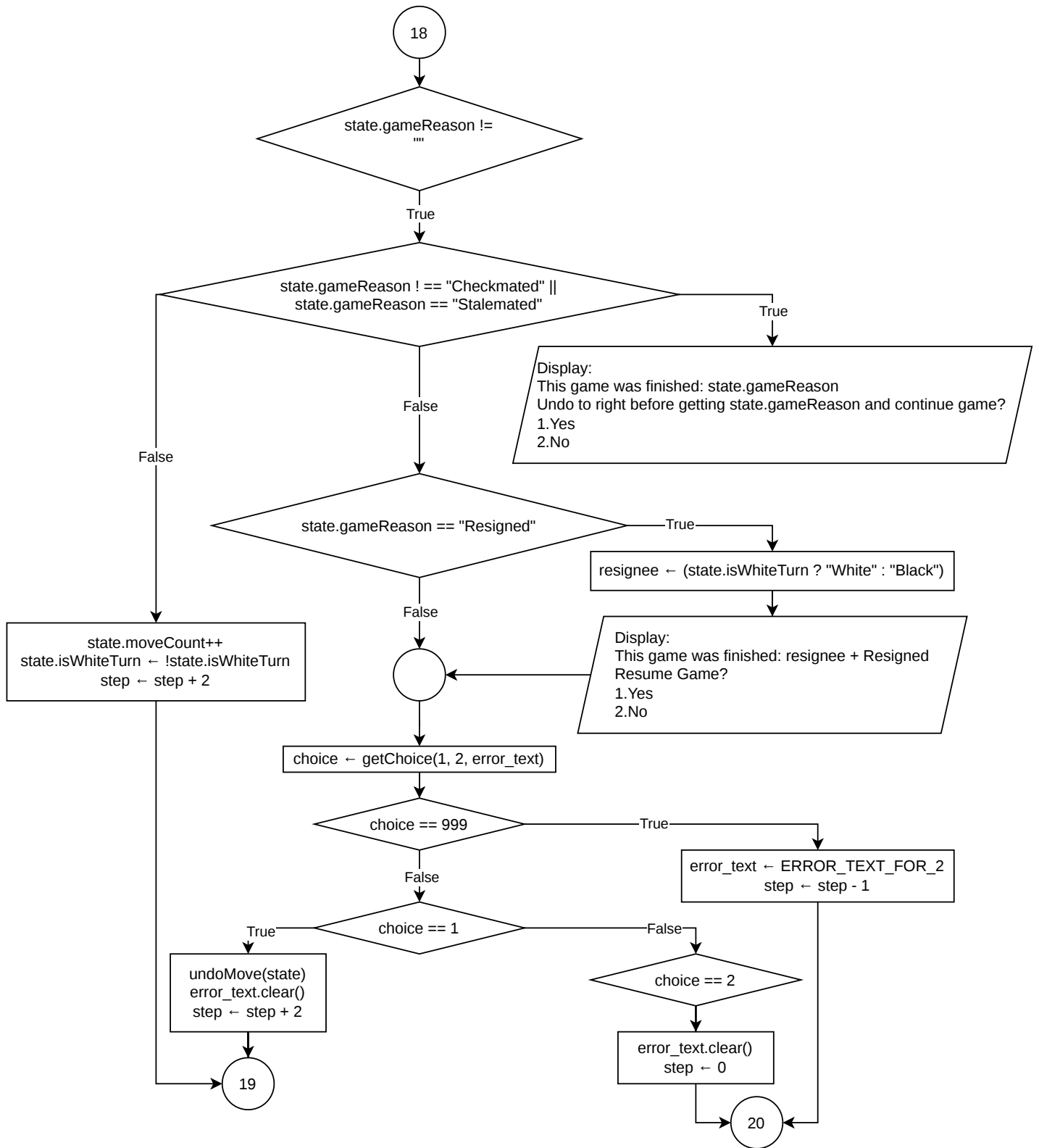


5.2.17 continueGame

This function lets user select a previous saved match to be continued, loads the state of game, and resumes the match loop.

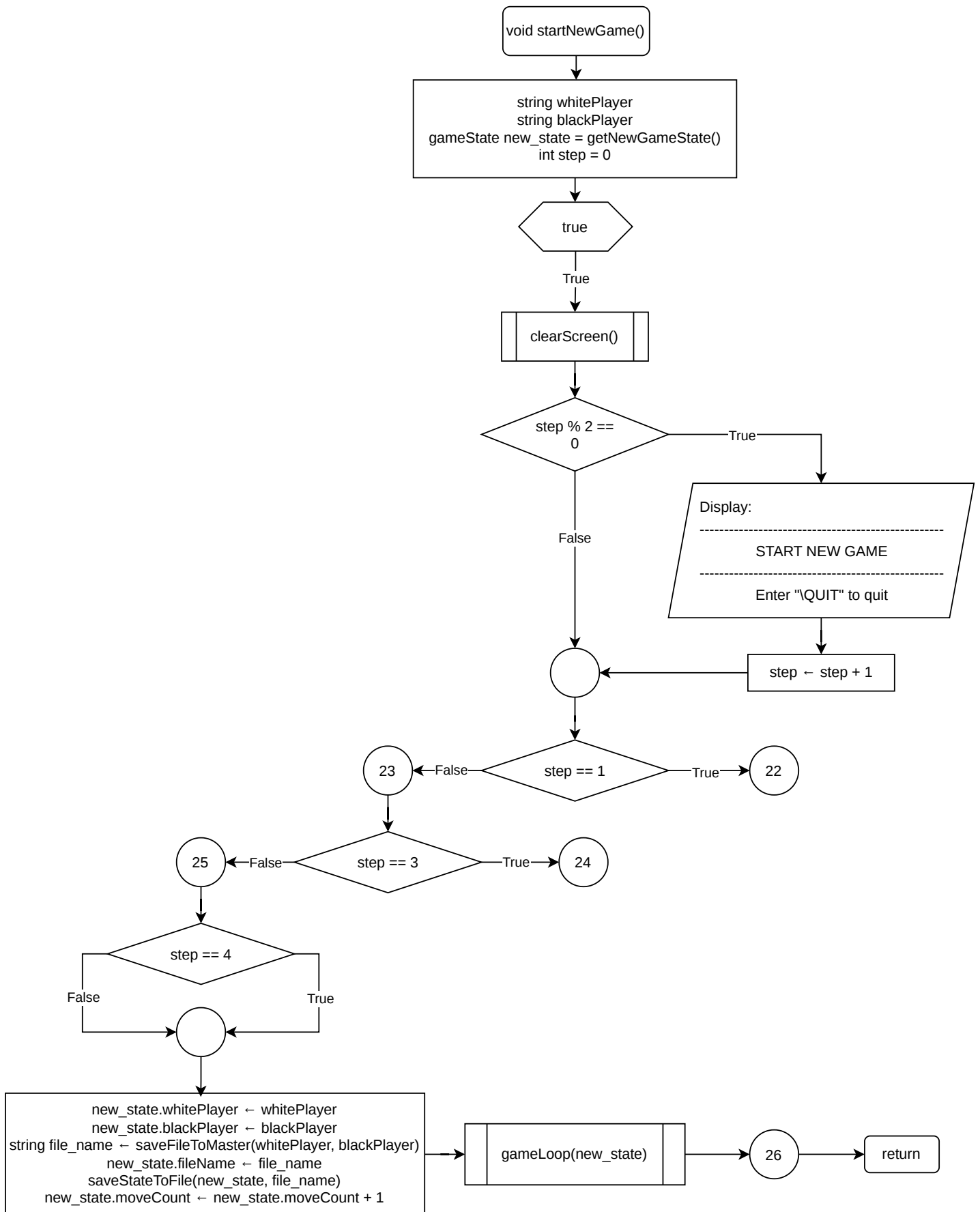


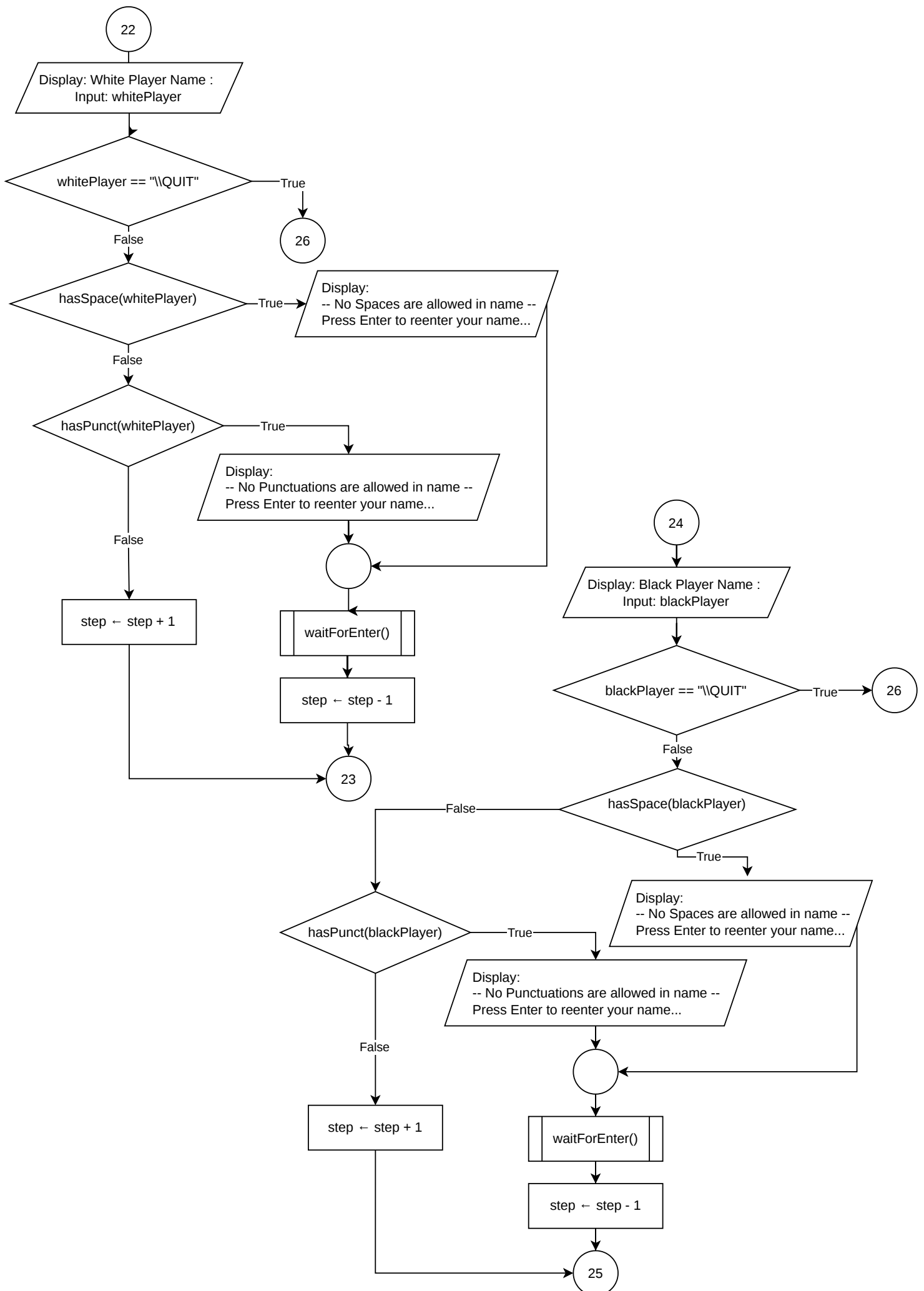




5.2.18 startNewGame

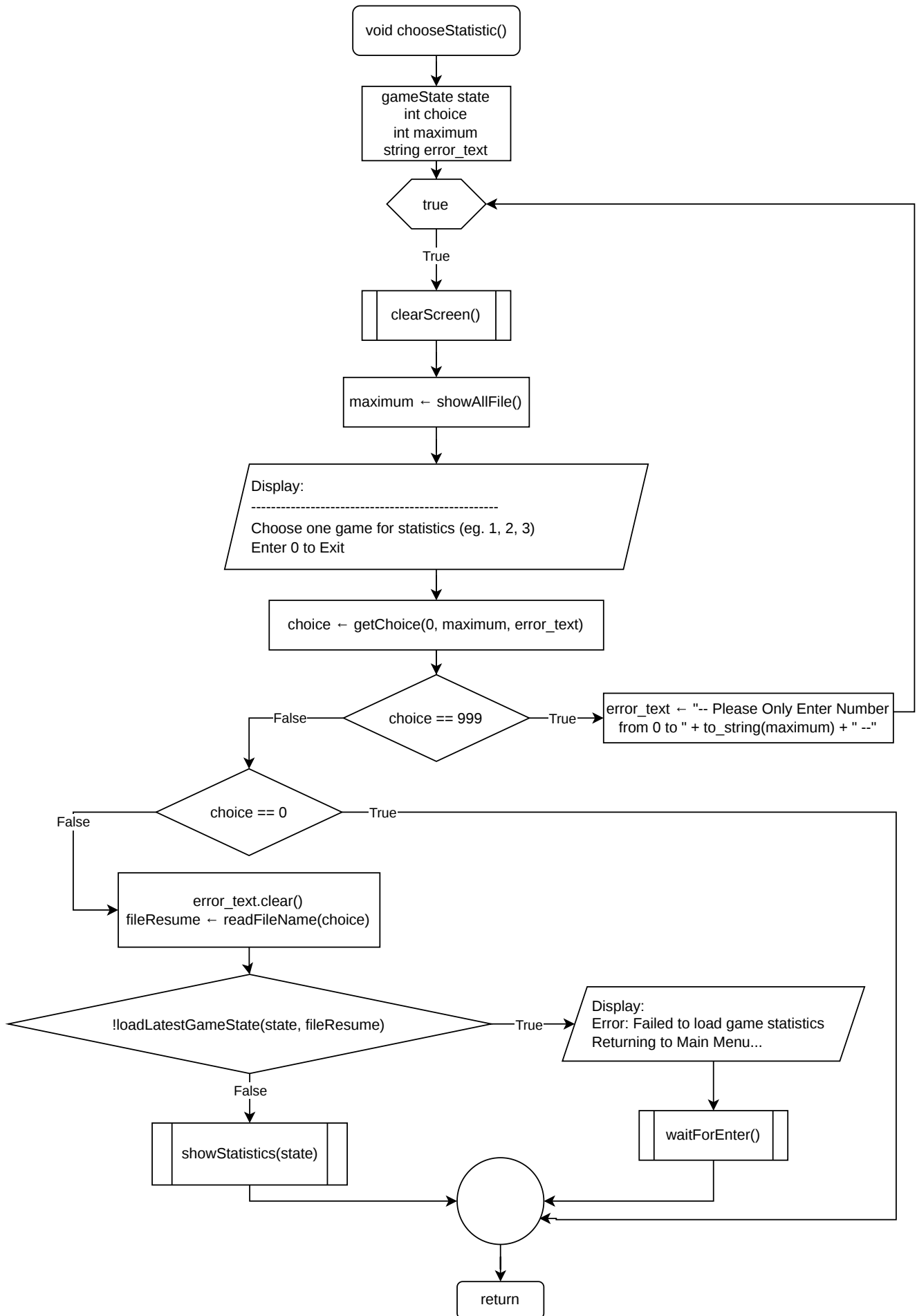
This function collects players names, creates a new game state and new save file, and starts a new game.





5.2.19 chooseStatistic

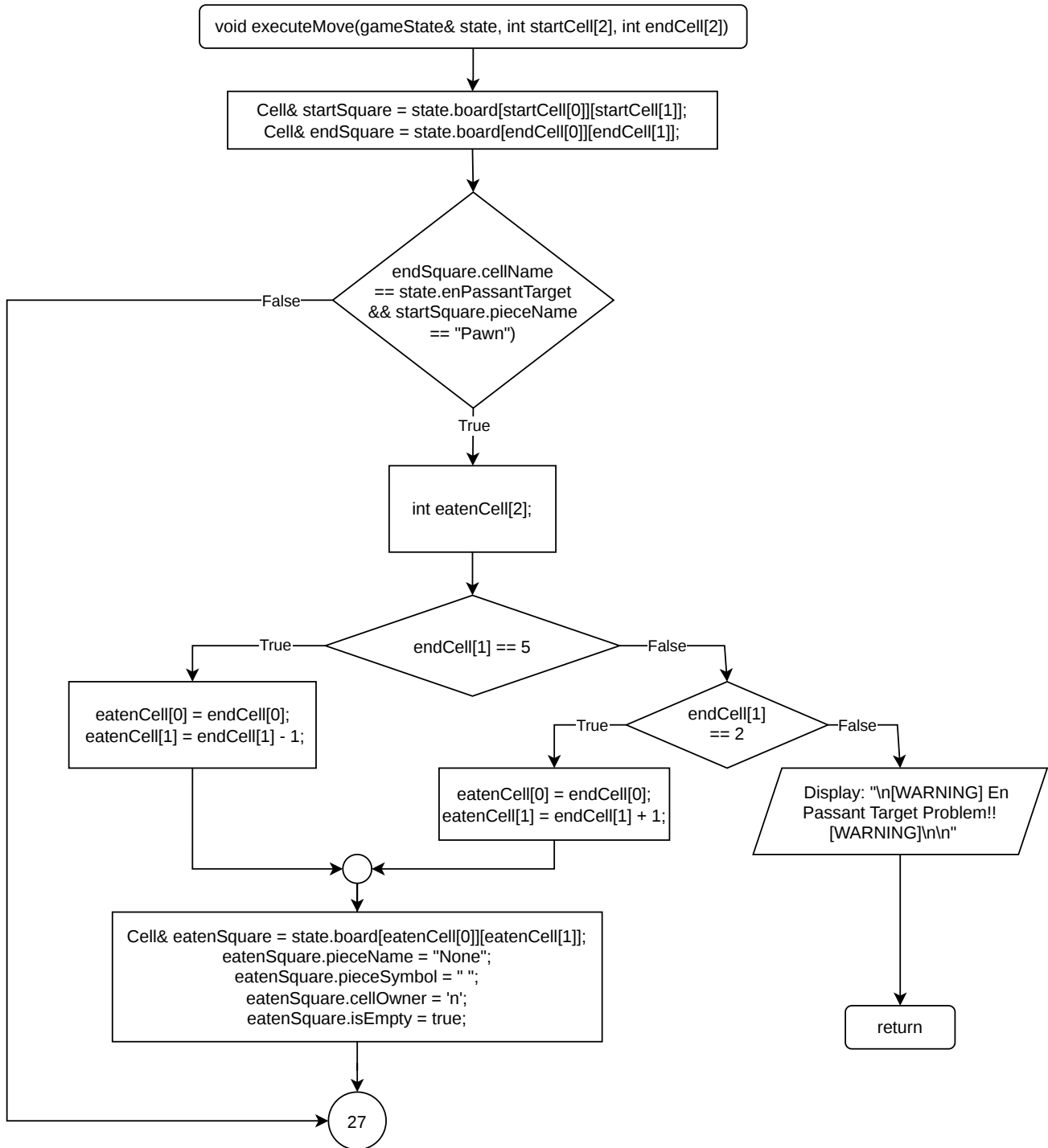
This function collects players names, creates a new game state and new save file, and starts a new game.

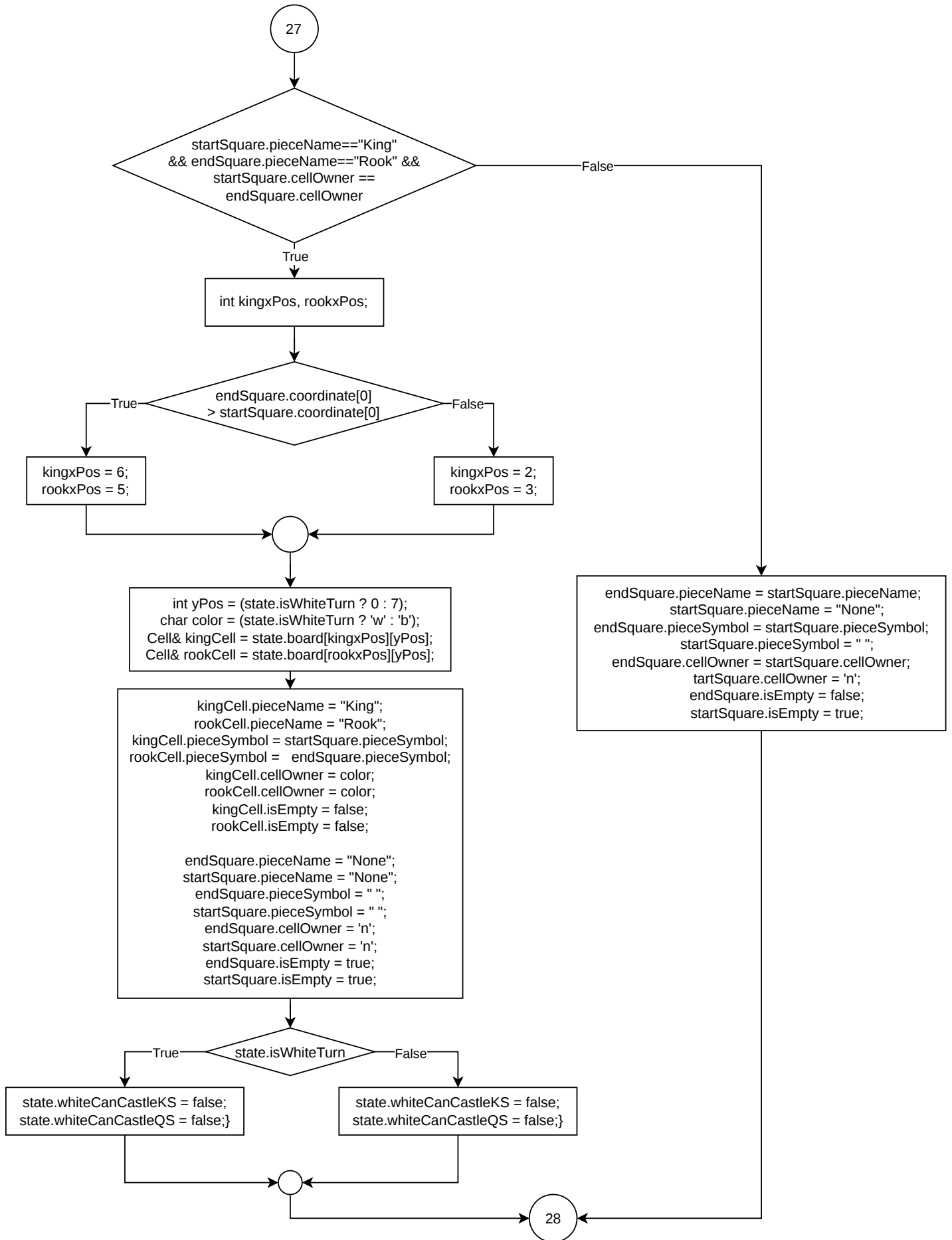


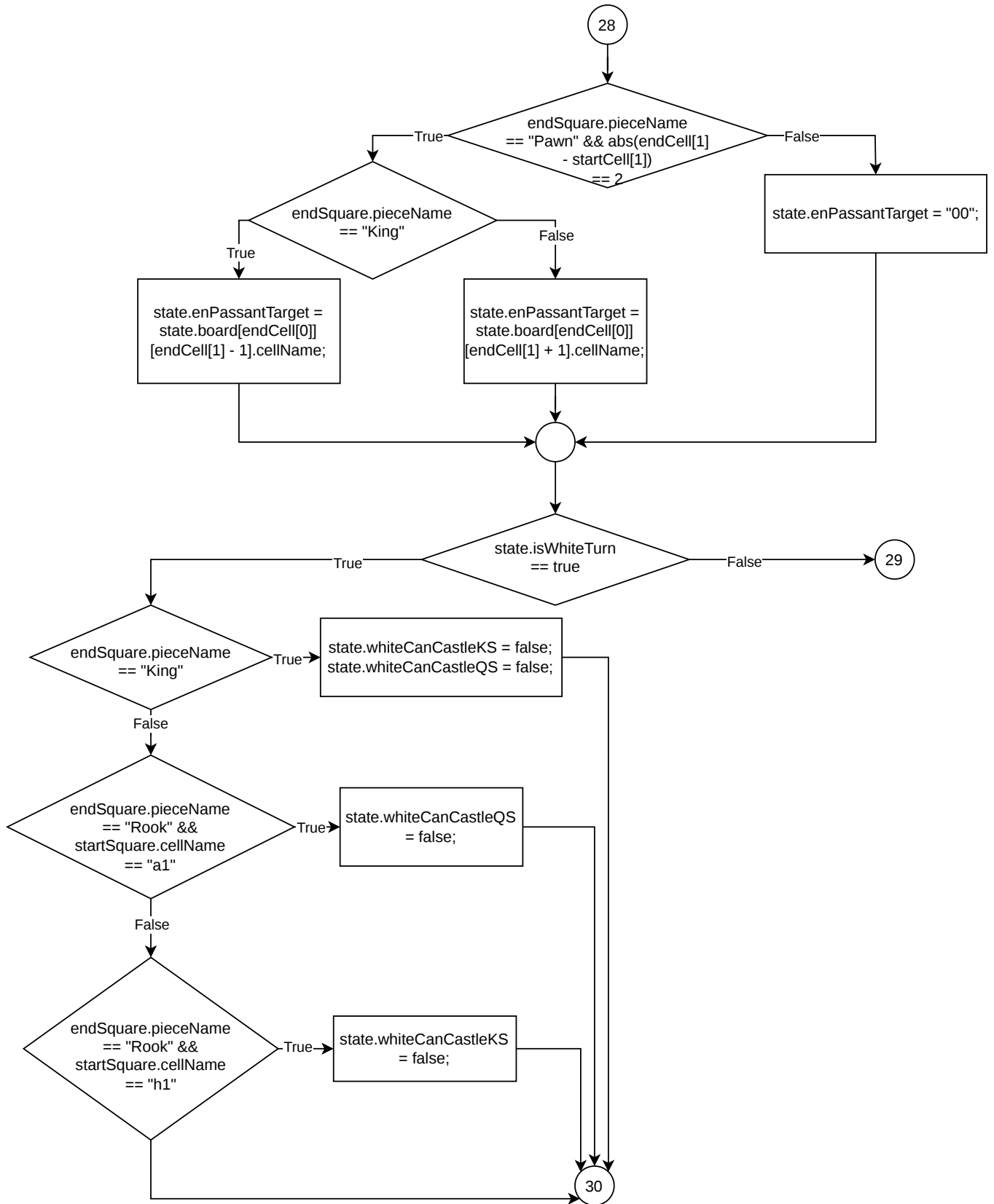
5.3 Game Logic

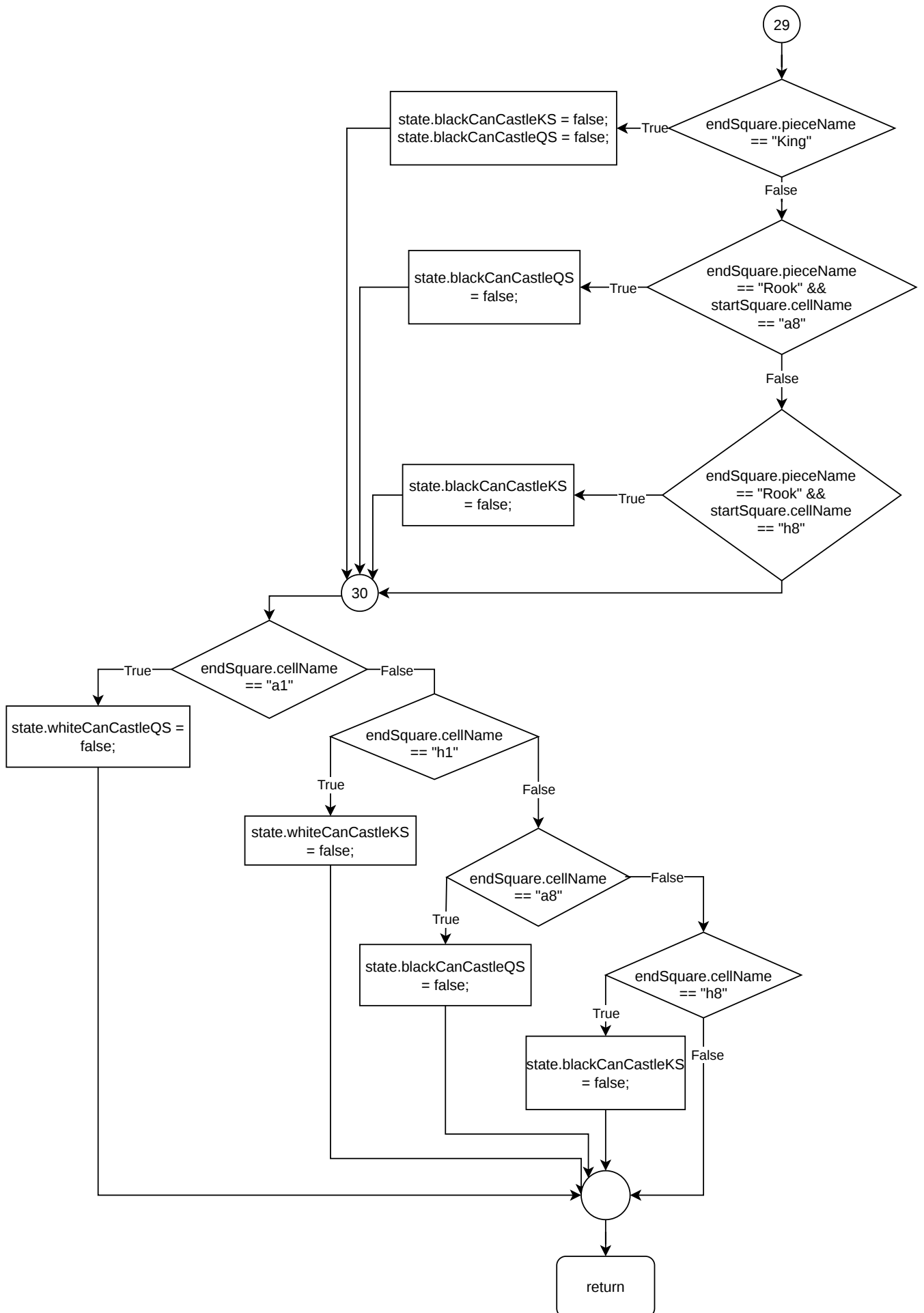
5.3.1 void executeMove()

This function is use to move a piece form one cell to another cell without validation. Special moving such as Castling and Empassant are included



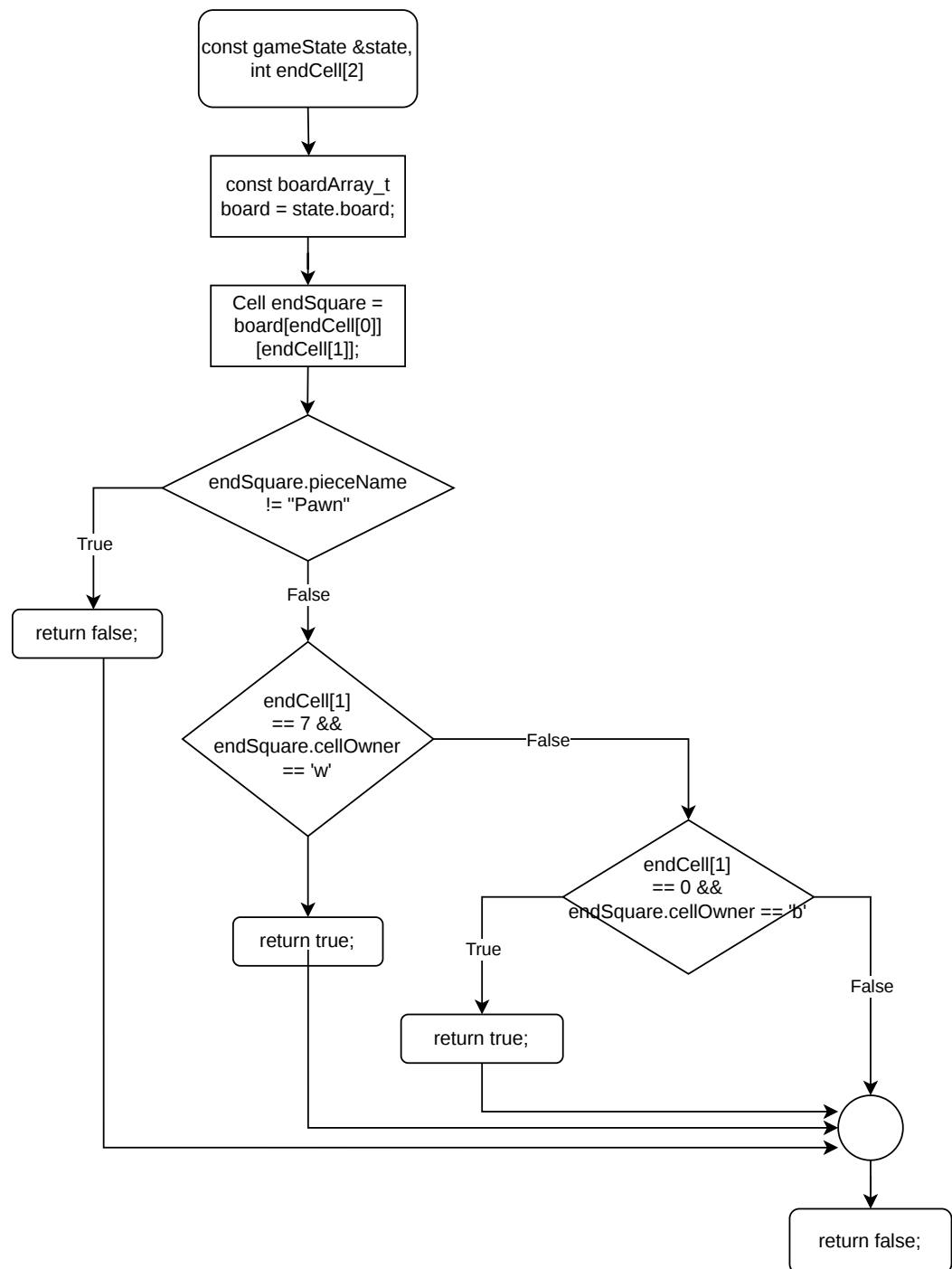






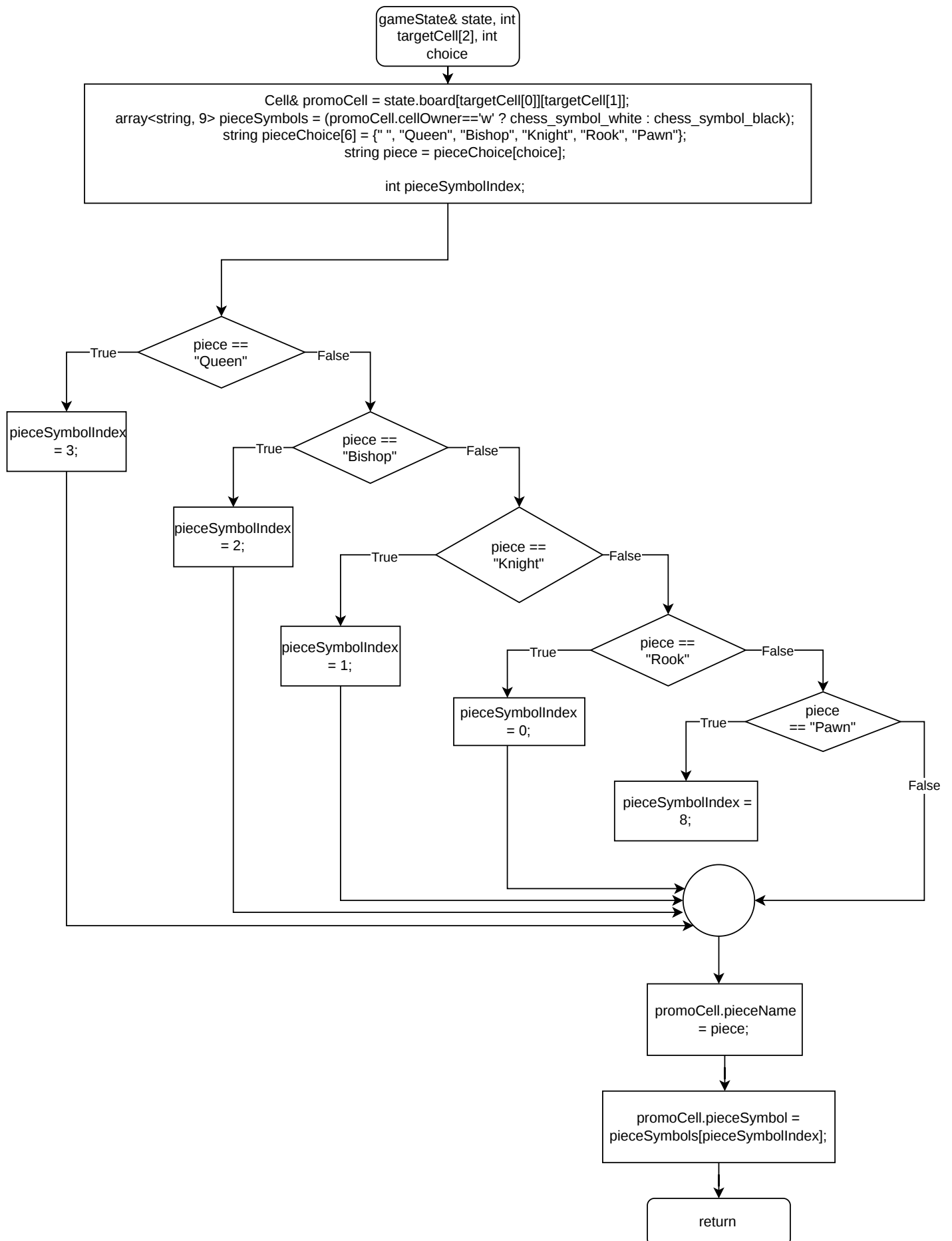
5.3.2 isPawnPromotion

The isPawnPromotion() function is to check whether a pawn has reached the opponent's end of the chessboard and is eligible for promotion, returning true if the pawn has reached the correct row and false.



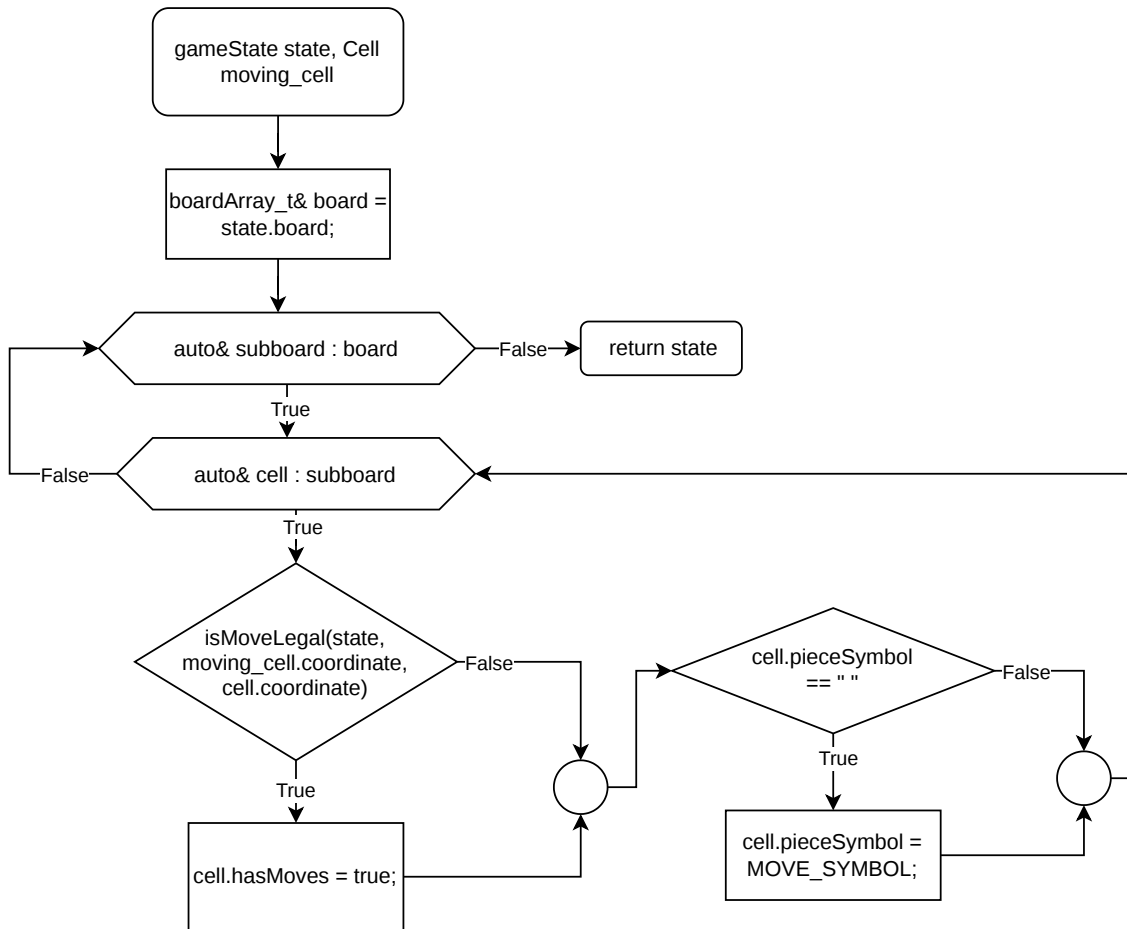
5.3.3 promotePawn

The promotePawn() function promotes a pawn to the chess piece selected by the player such as Queen, Bishop, Knight or Rook and updates the pawn's piece name and symbol accordingly.



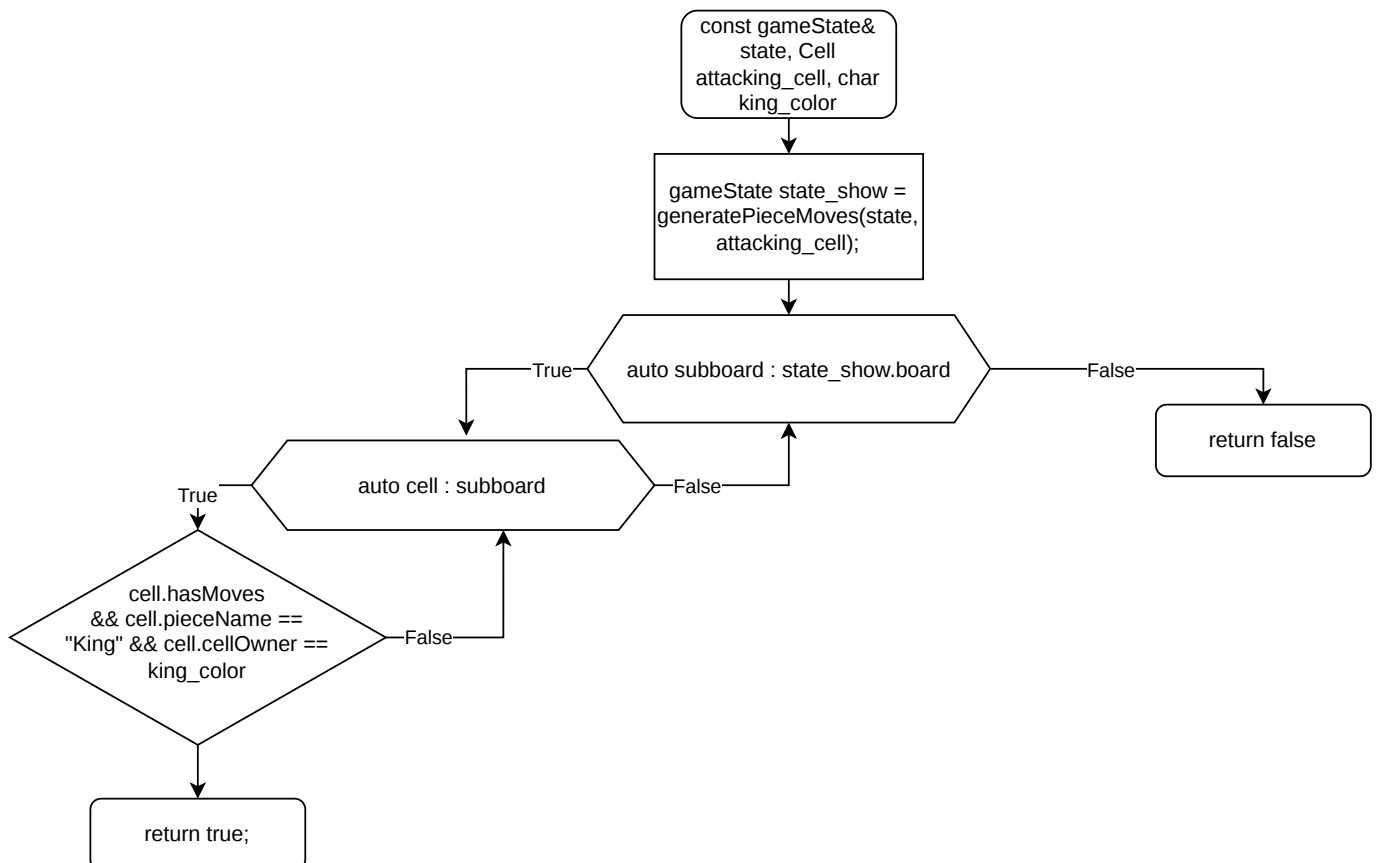
5.3.4 generatePieceMoves

The generatePieceMoves() function generates and marks all possible moves for a selected chess piece on the board, including moves that may leave player's own king in check, by updating the corresponding cells with a move indicator before returning the updated game state.



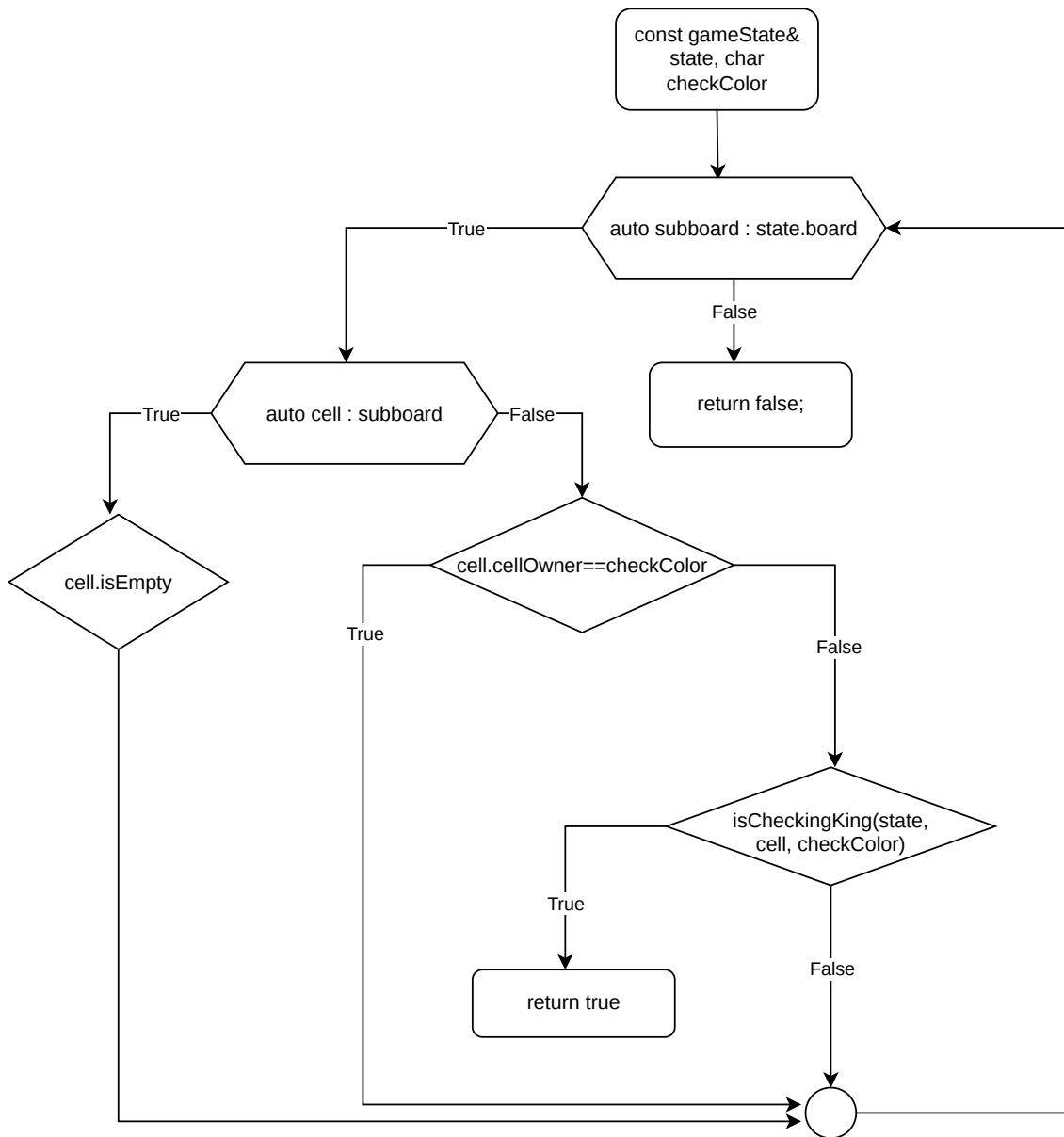
5.3.5 bool isCheckingKing

The isCheckingKing() function is to check whether a specific chess piece is currently able to attack the king of a specified color and returns true if the king is being checked or false otherwise.



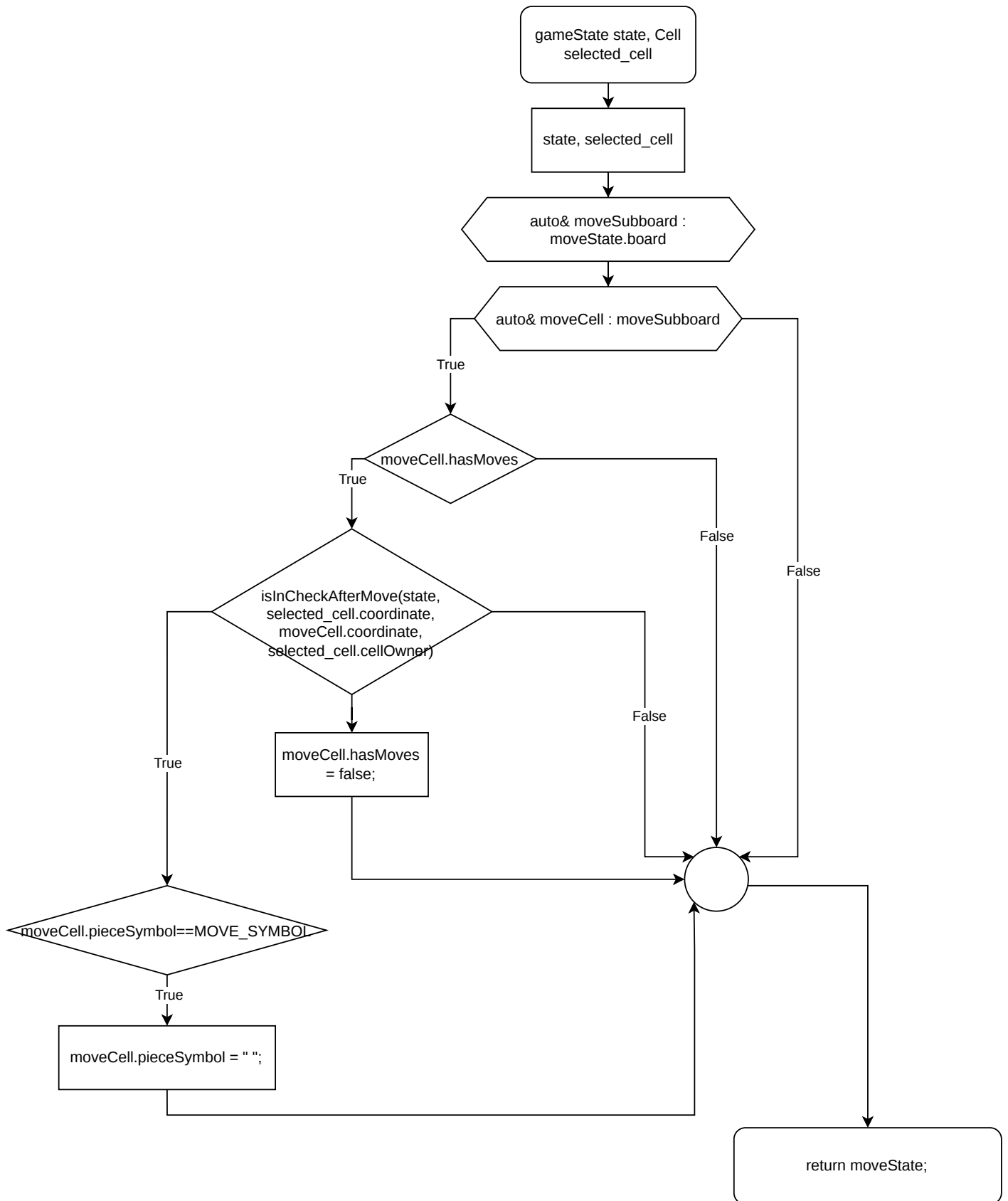
5.3.6 bool isInCheck

The `isInCheck()` function checks whether the king of a specified color is currently in check by an opponent's piece, returning true if the king is being and false if it is not.



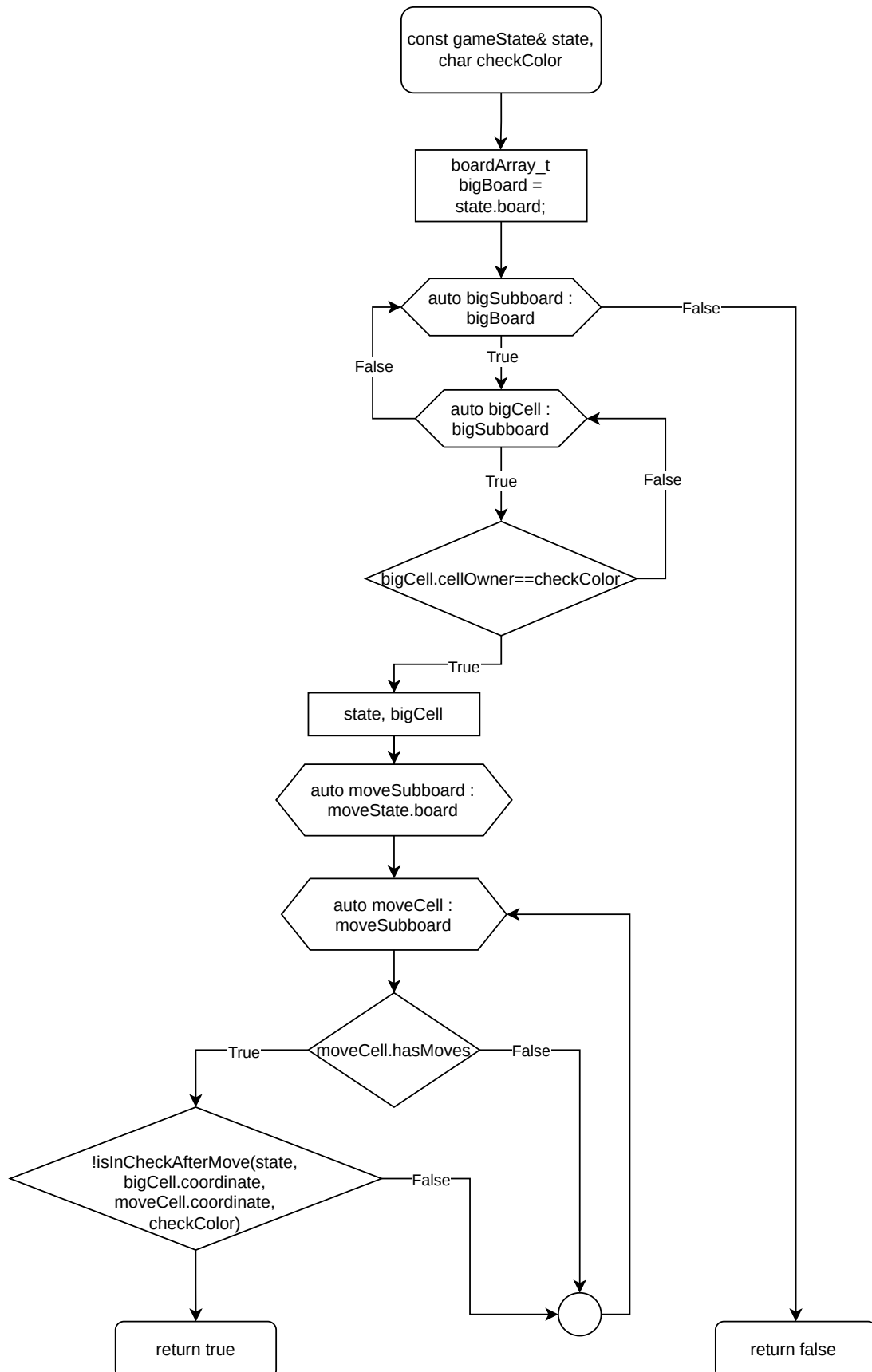
5.3.7 gameState show AvailableMoves

The show AvailableMoves() function determines and displays all legal moves for a selected chess piece by removing any moves that would leave the player's own king in check, ensuring that only safe and valid moves remain.



5.3.8 bool hasLegalMoves

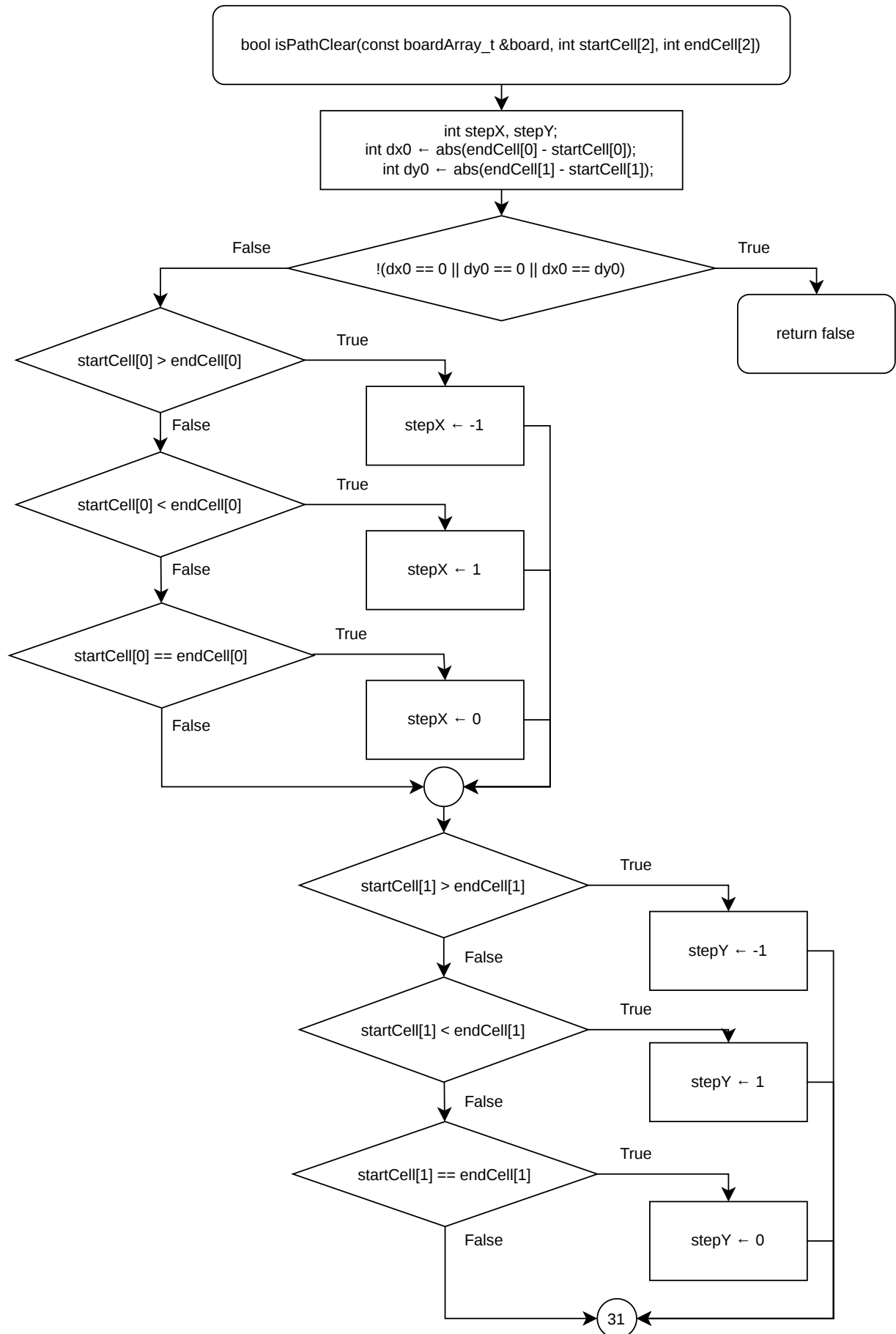
The hasLegalMoves() function checks whether a player has at least one legal move that does not leave their own king in check, returning true if a legal move exists and false if no legal moves are available.

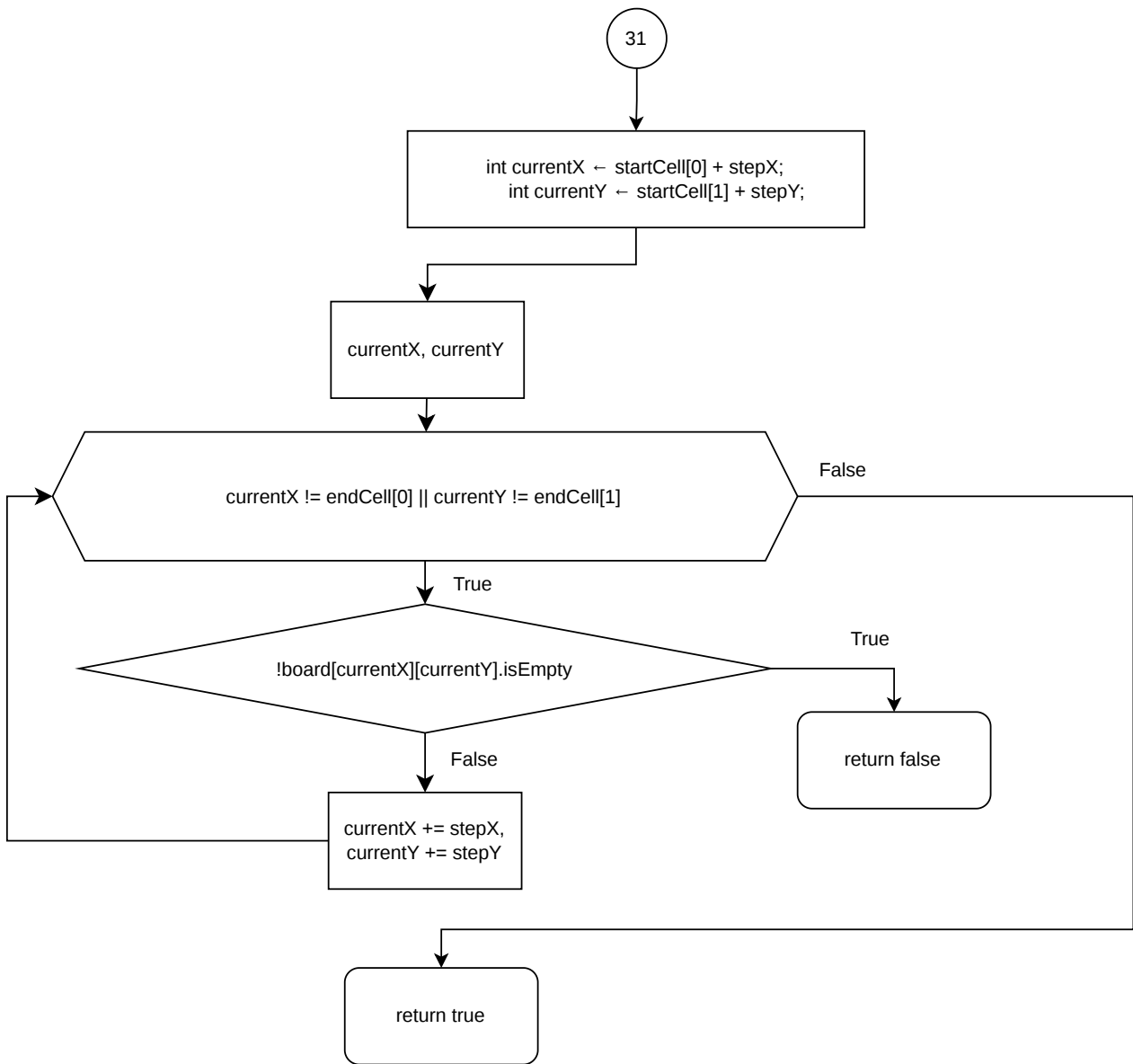


5.4 Move Validation

5.4.1 isPathClear

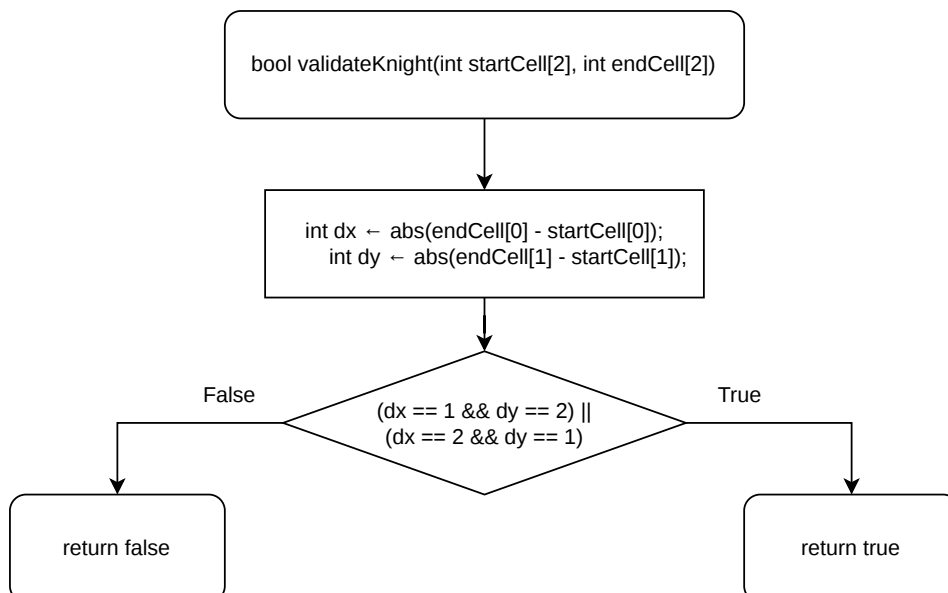
This function identifies the increasing or decreasing for both row and column movement first, then tests all the pathway one by one until found a non-empty square





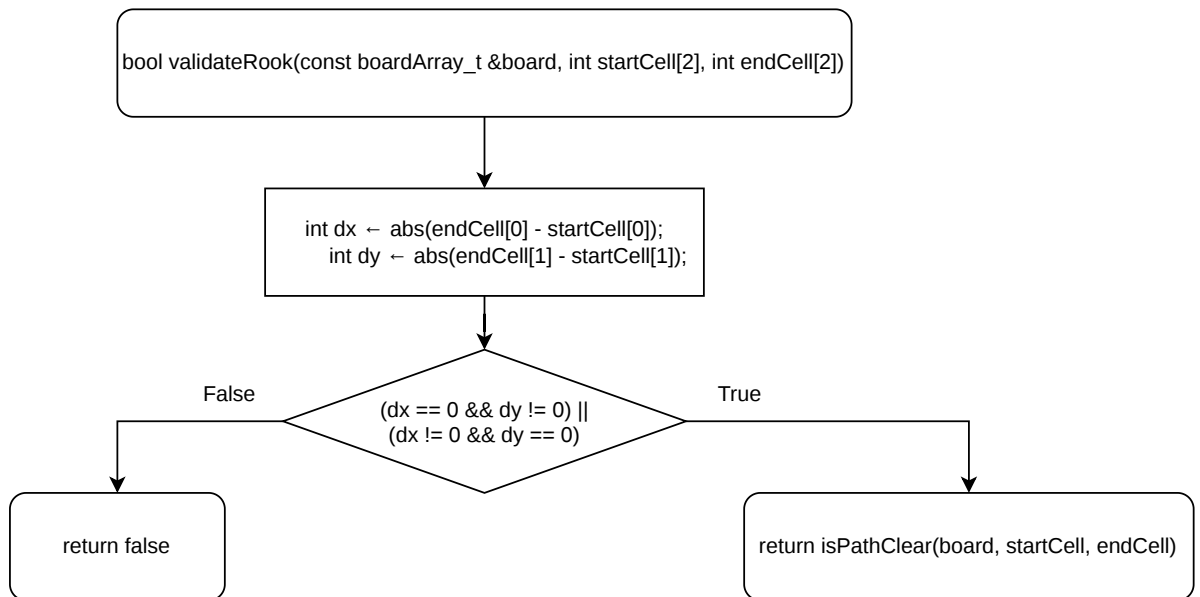
5.4.2 validateKnight

This function validate the movement of Knight, which is either move 2 step in row and move 1 step in column, or move 1 step in row and move 2 step in column



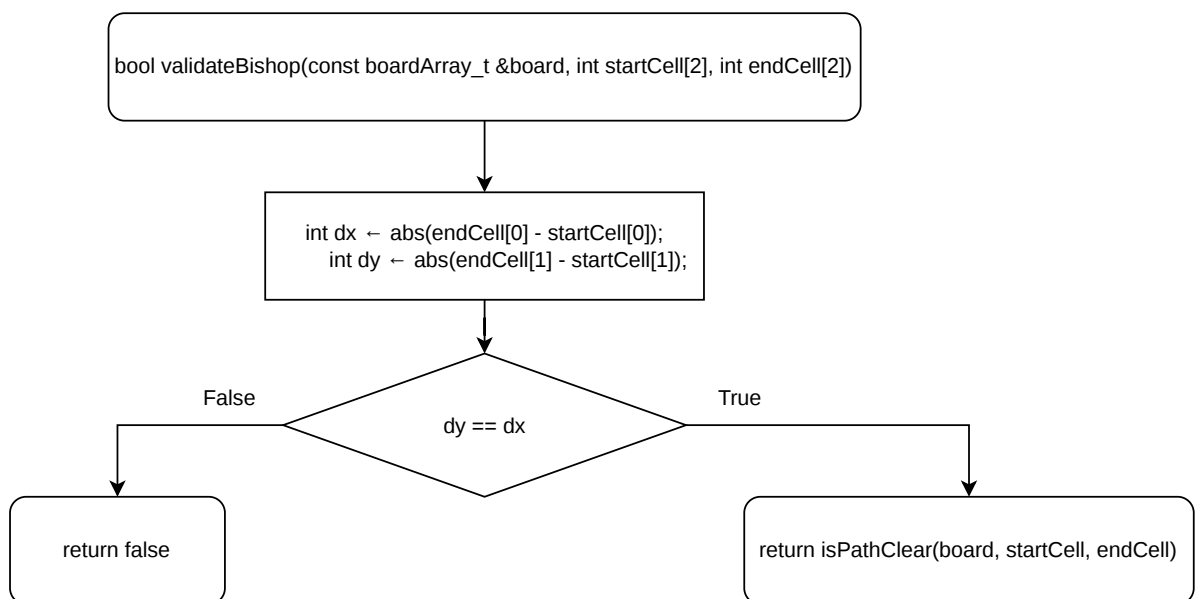
5.4.3 validateRook()

This function validate the movement of Rook, which is either no move in row but move in column, or move in row but no move in column



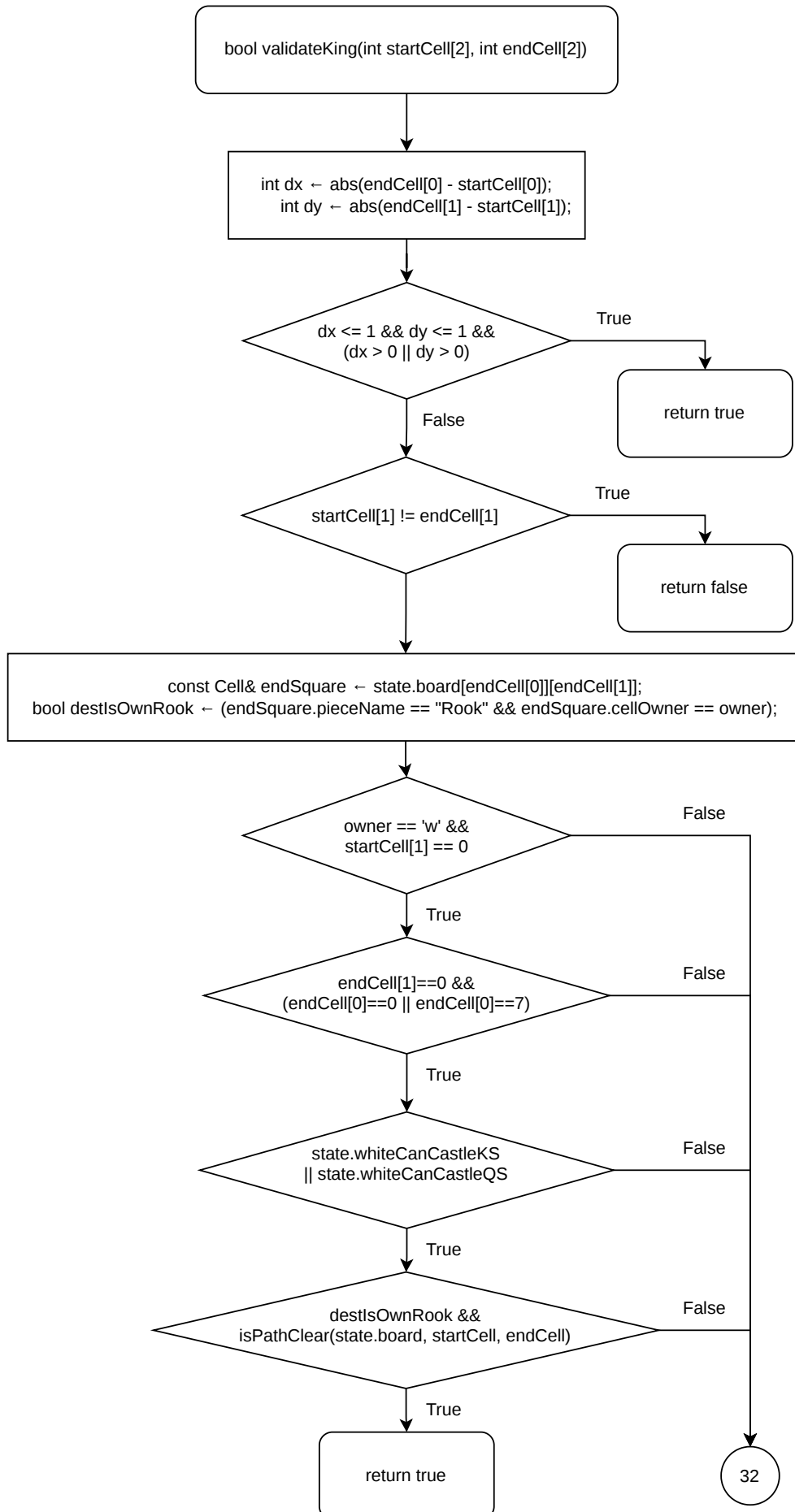
5.4.4 validateBishop()

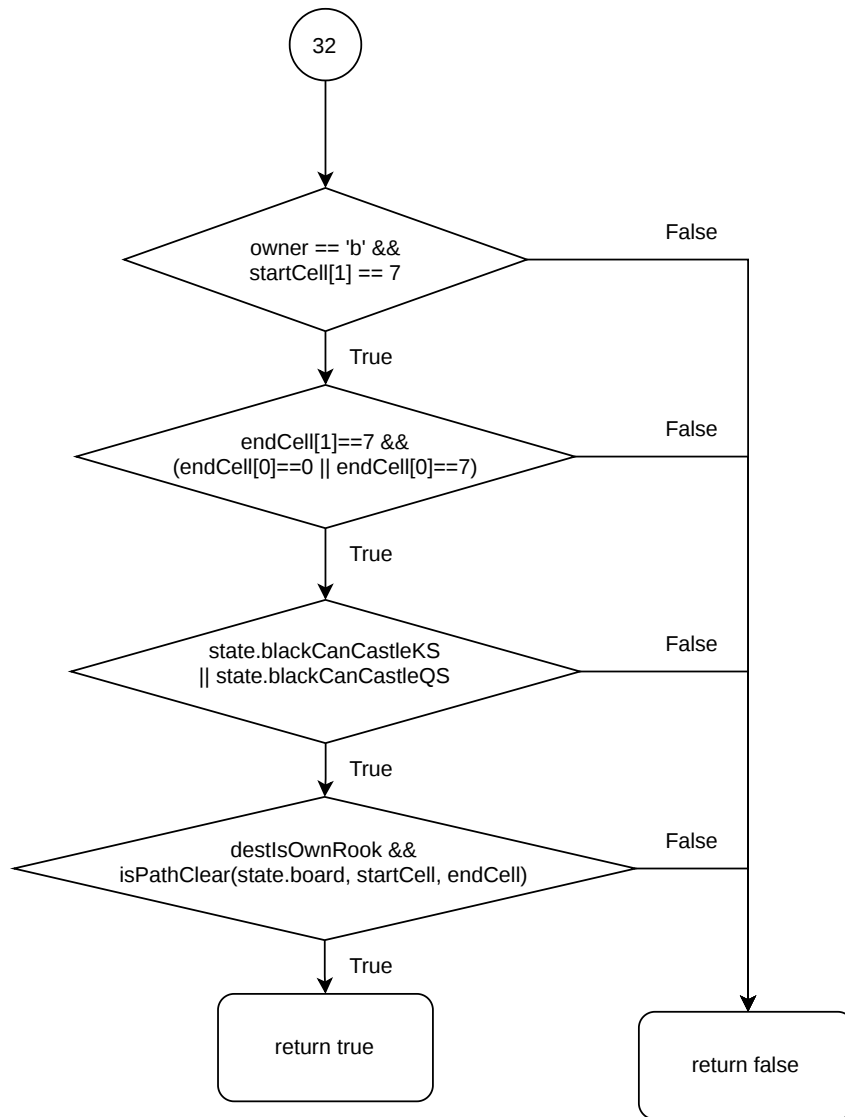
This function validate the movement of Bishop, which is the move in row and column are same



5.4.5 validateKing

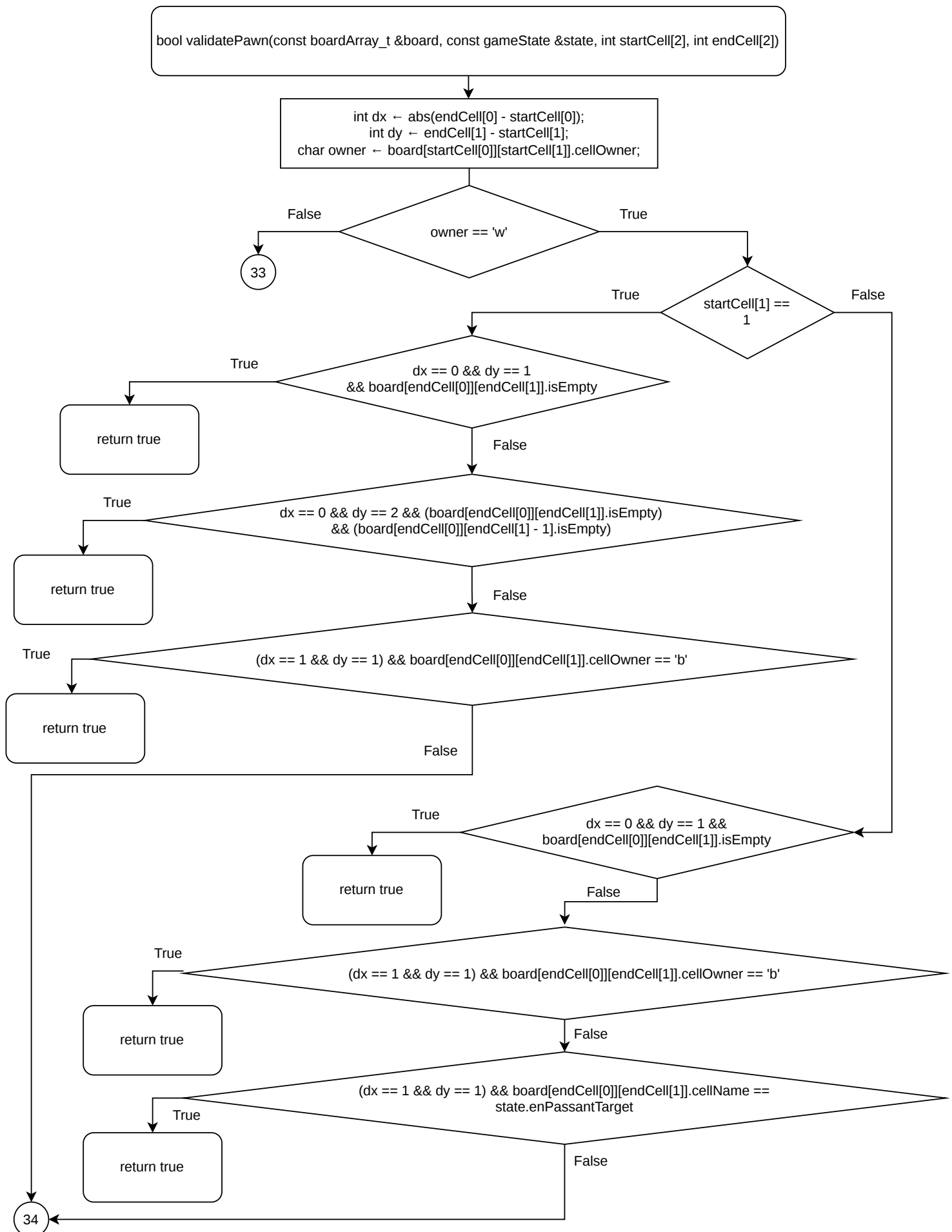
The function validate the movement of King, which is both the move in row and column should not be more than 1. But if the King and the Rook is still in their start square, casting is allowed.

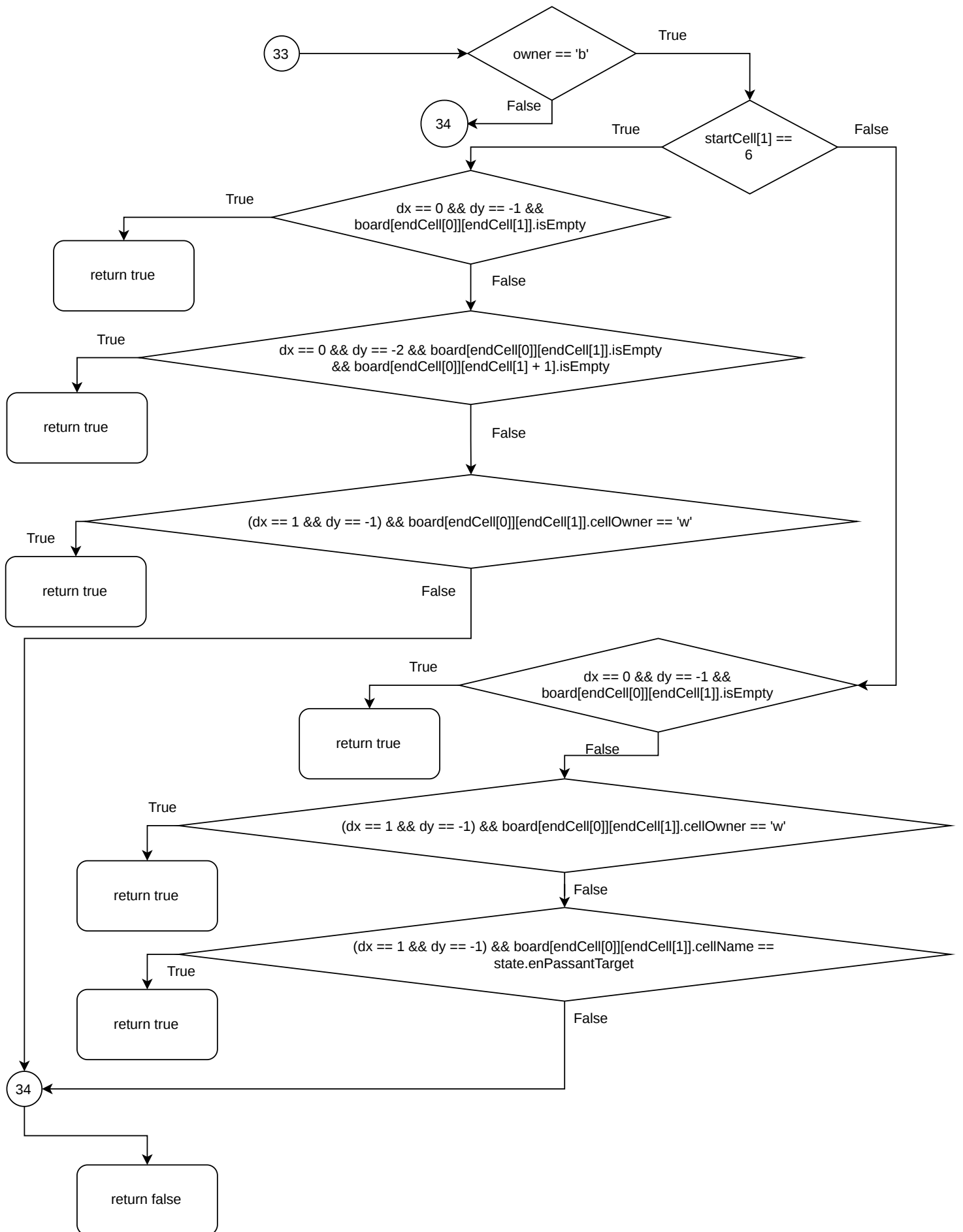




5.4.6 validatePawn

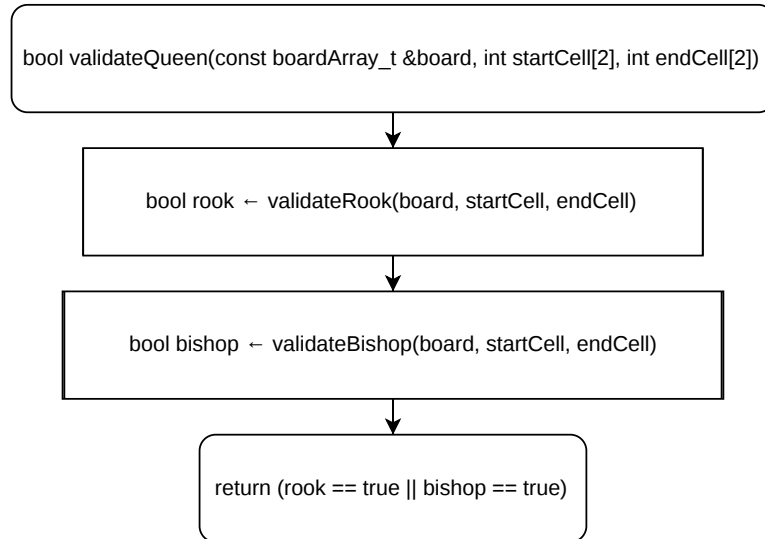
This function validate the movement of Pawn, which is both side can only move forward 1 step or 2 step if first move. If there is a capture for Pawn, the Pawn can also move forward 1 step and move to left or right 1 step.





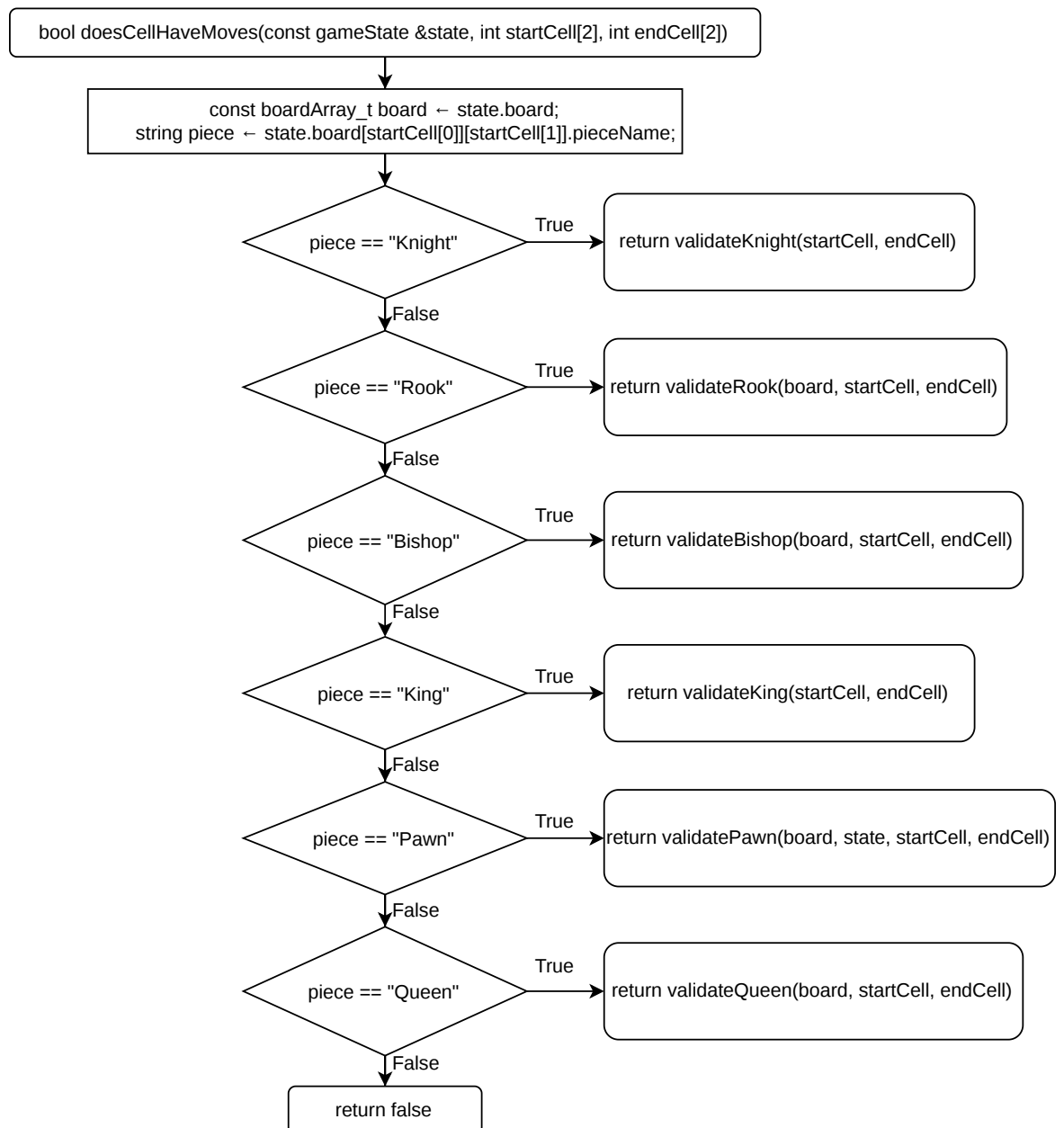
5.4.7 validateQueen

This function validate the movement of Queen, which is just validate as Rook and Bishop



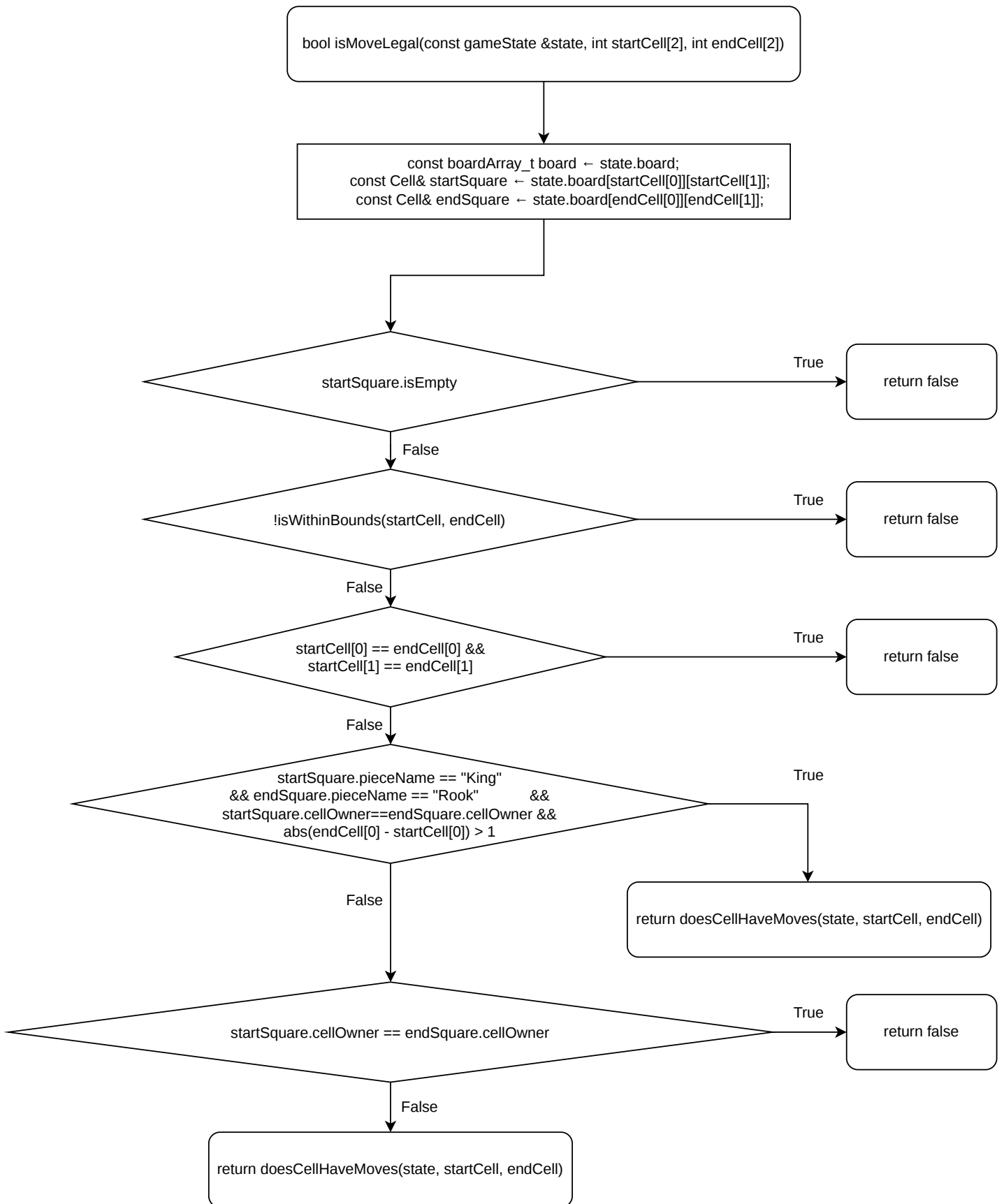
5.4.8 doesCellHaveMoves

This function check the type of the piece, and then validate the movement of each type of piece by each validation function



5.4.9 isMoveLegal

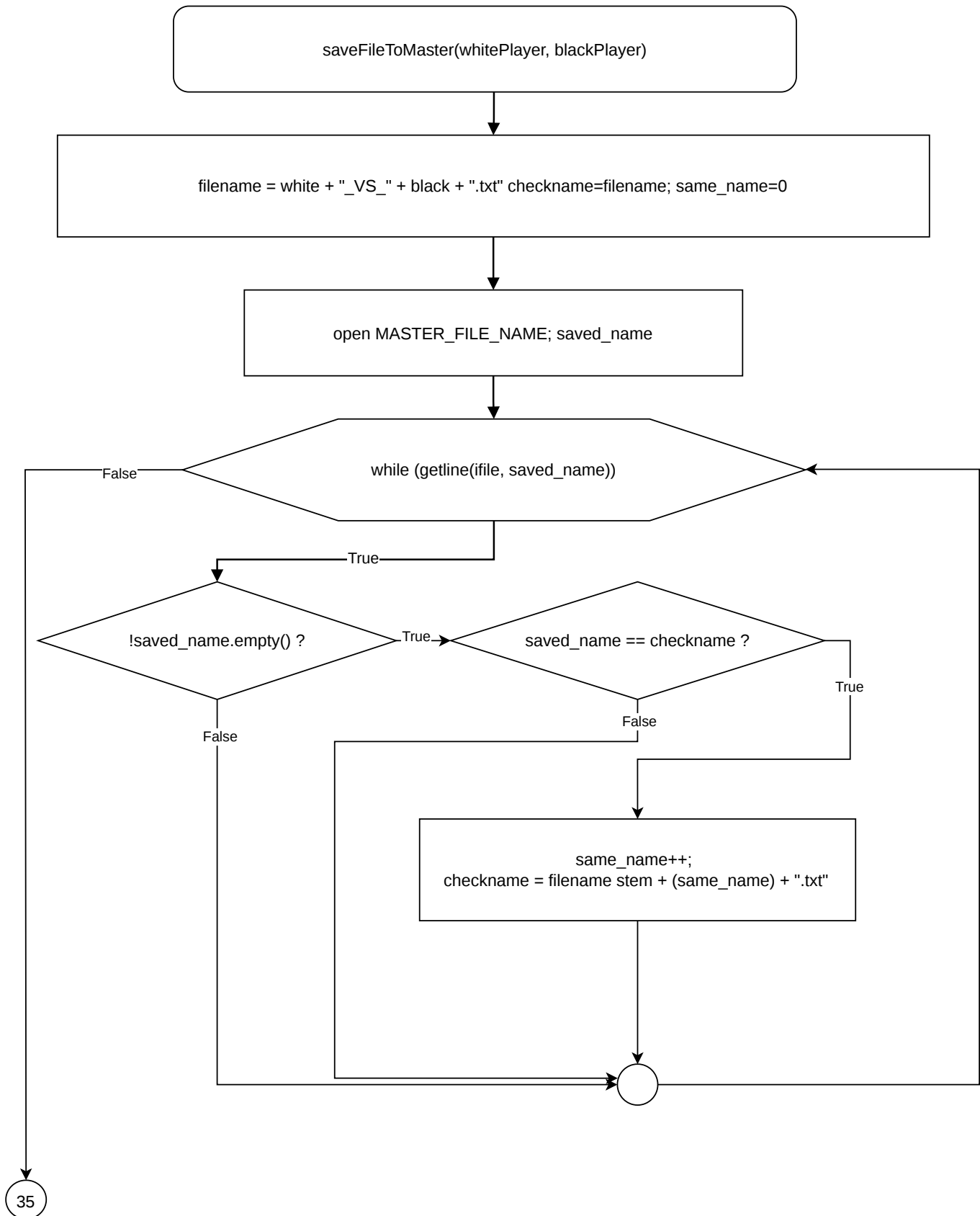
This function check either is the current player turn's piece been selected.

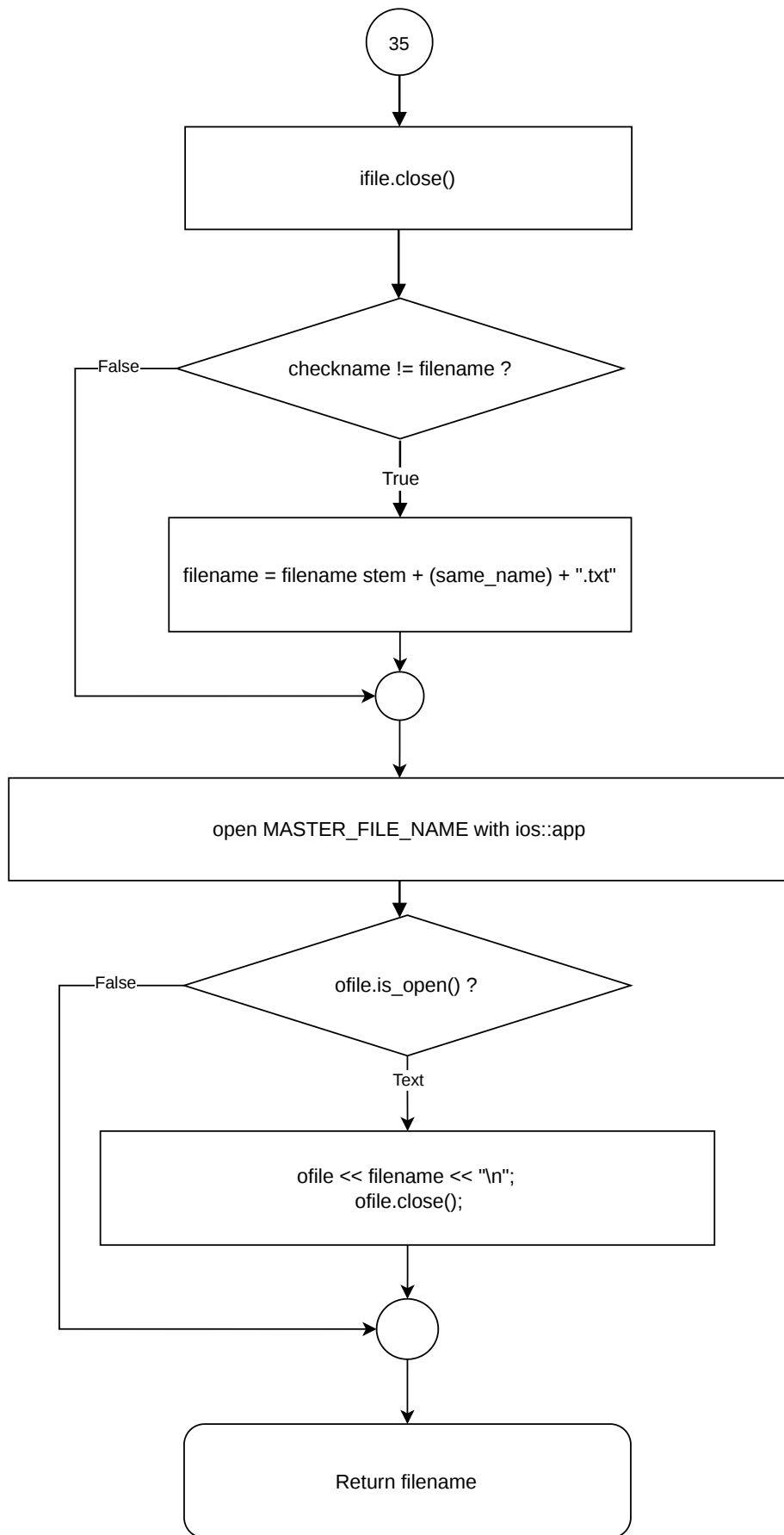


5.5 Storage

5.5.1 saveFileToMaster

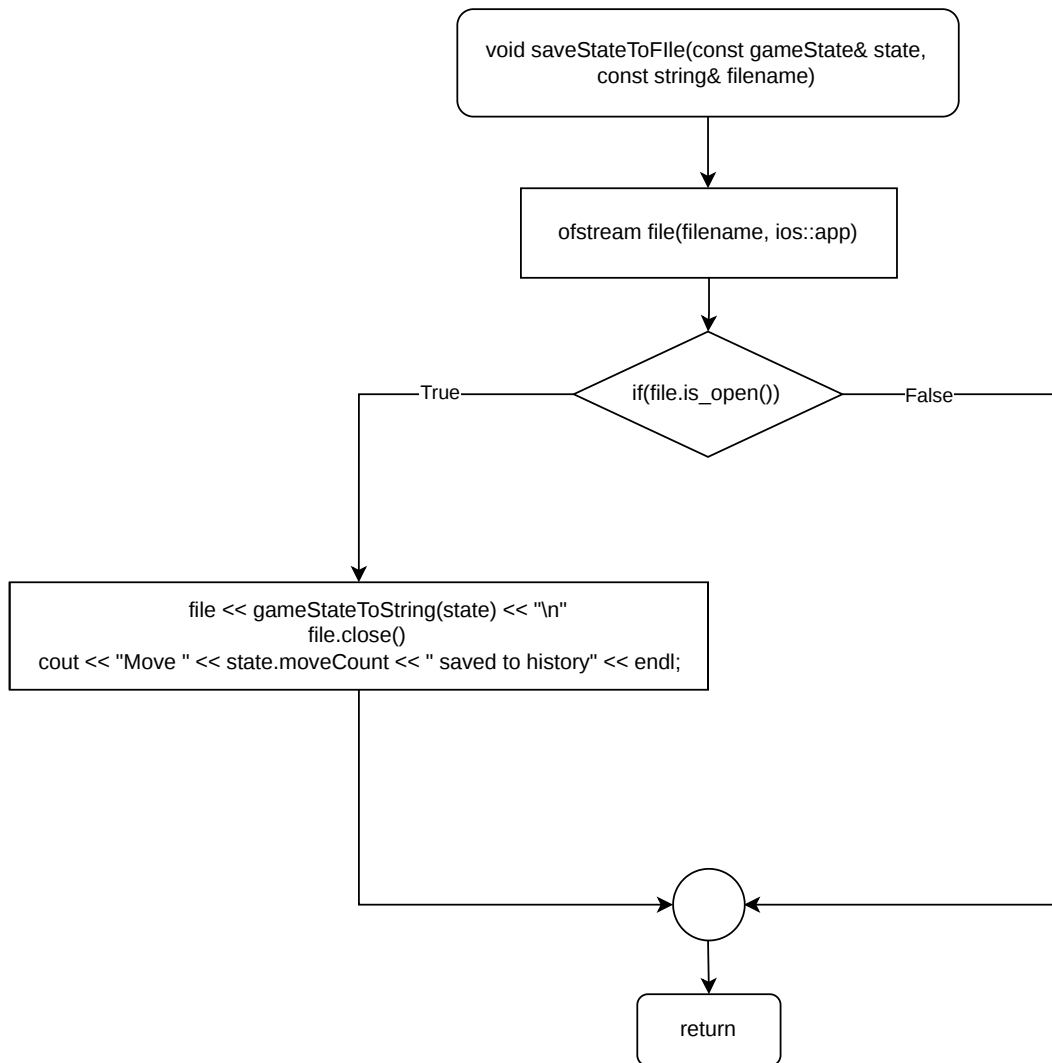
This function generate the filename using the player names first, then check whether the filename already exist in the master file. If the filename does not exist, it will append the filename into the master file and return the filename after checking the file can be opened successfully.





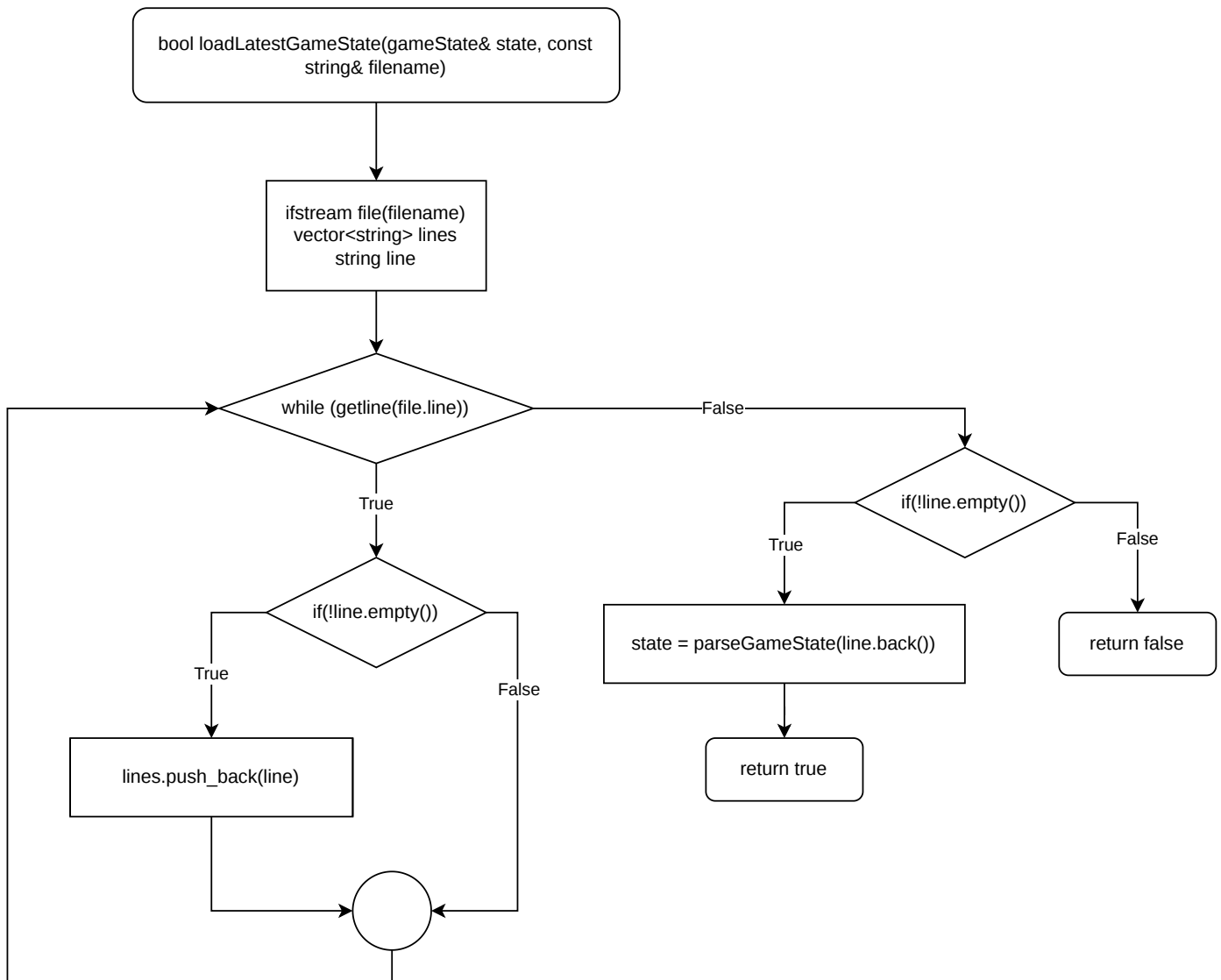
5.5.2 saveStateToFile()

This function opens the specified .txt file in append mode and saves the current game state to the file without overwriting previously stored game data. After successfully saving the game state, the file is closed and a confirmation message displaying the saved move number is shown to the user.



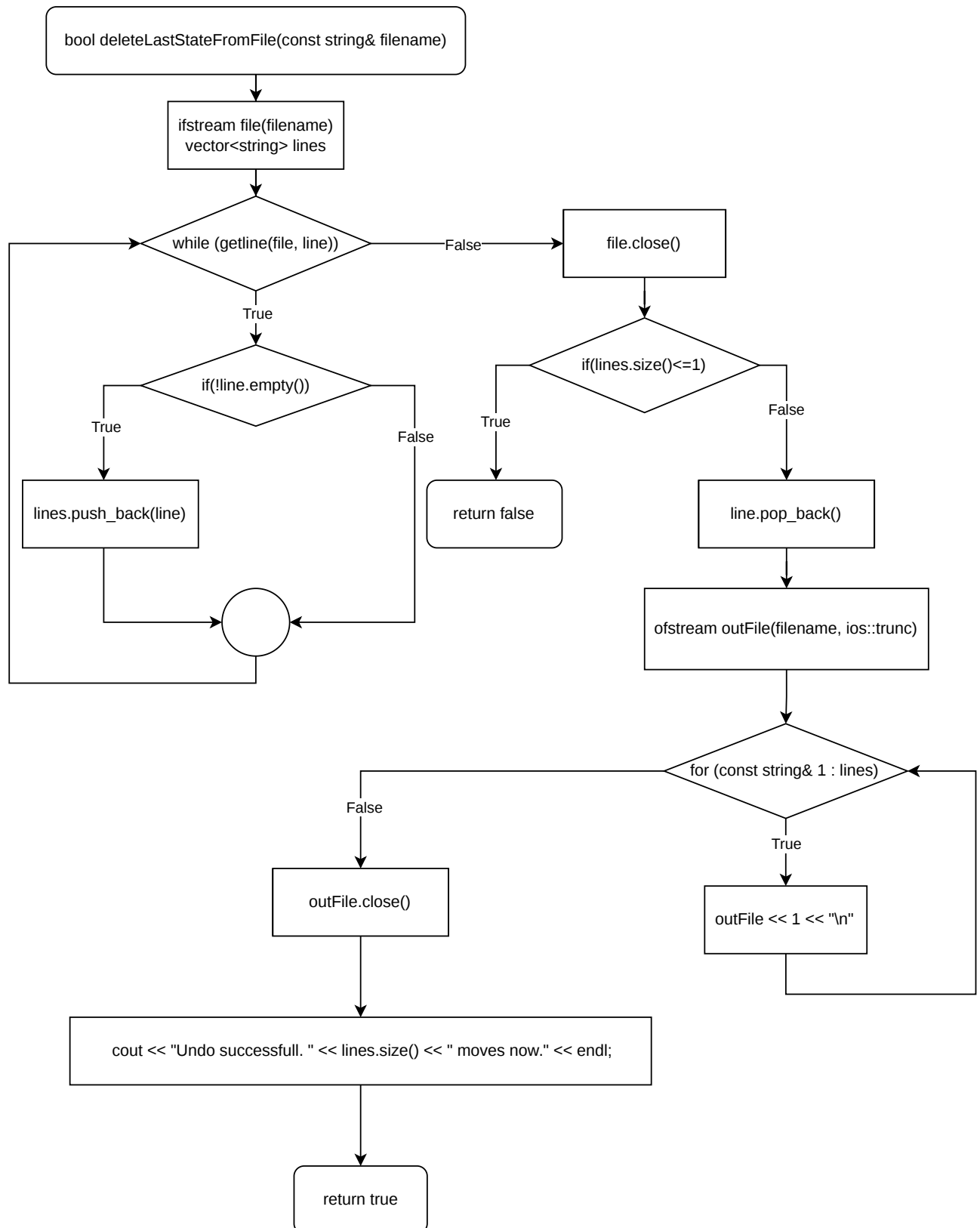
5.5.3 loadLatestGameState()

The function opens the specified .txt file and reads the saved game states line by line. Non-empty lines are stored in a list, and the latest saved game state is obtained from the last line of the list. The retrieved game state is then parsed and loaded into the current game state. The function returns true if a saved game state is successfully loaded and false if the file contains no saved game data.



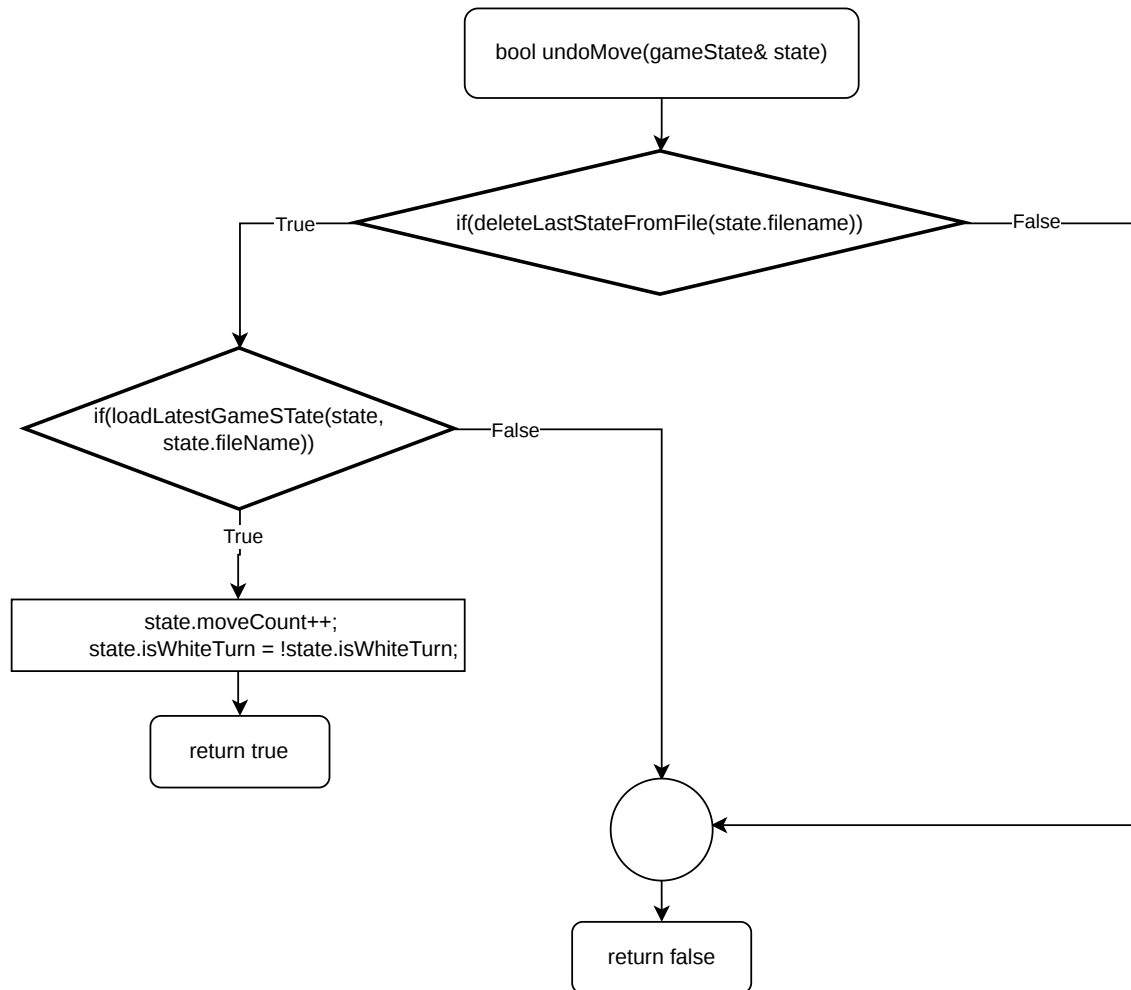
5.6.4 bool deleteLastStateFromFile()

The function opens the specified .txt file and reads all non-empty saved game states into a list. If the file contains no saved game states, the function returns false. Otherwise, the latest saved game state is removed from the list. The remaining game states are then written back to the .txt file using truncation mode, which replaces the previous file contents with the updated list. After successfully updating the file, a confirmation message showing the remaining number of moves is displayed, and the function returns true.



5.5.5 undoMove()

The function performs the Undo Move operation by first deleting the latest saved game state from the specified .txt file. If the deletion is successful, the function loads the new latest game state from the file. If both operations are successful, the function returns true to indicate that the move has been successfully undone. Otherwise, it returns false.



5.5.6 int showAllFile()

The function opens the master .txt file containing the saved game records and reads the file line by line. For each non-empty line, it displays a numbered game record in the Game History section. The last four characters of each filename are removed before displaying the game name. After all records have been displayed, the function closes the file and returns the total number of saved games.

